THE SPECTRE KNOT: ONE ENDLESS STRAND ON AN APERIODIC MONOTILE

JUSTIN ROUGHLEY

Abstract. The Spectre of Smith, Myers, Kaplan and Goodman-Strauss tiles the plane, but only non-periodically. We decorate its straight-edged form Tile(1, 1) with fourteen ribbon pieces, the same on every tile, so that ribbons continue straight across every tile edge. We prove that the pieces never close up: in every tiling by copies of Tile(1, 1) of one handedness, including tilings with faults, the ribbons contain no closed strand (the Endless Strand Theorem). In the nest tiling, grown from a single tile by repeating the substitution so that each block is child 1 (a Δ-supertile) of the next, all the pieces of all the tiles join into a single strand, a two-way infinite path that passes through every tile (the Ariadne Theorem). We do not claim this for other tilings, and computations indicate that it fails in some. The proof shows that every supertile, at every level, connects the ribbon ends on its boundary in the same simple pattern: a hook at every boundary vertex but two, and one crossover. Where supertiles are glued, the ribbons zigzag along the shared boundary, so lengthening it by two edges changes nothing; and the side lengths at every level are reached from those at level 1 in steps of two, because every side length is a Fibonacci number, twice one, or one plus or minus two, with the same parity at every level. This reduces the infinite problem to finitely many cases at level 1, which we check by computer. The single strand is reduced in the same way to a map, computed at level 1, that says where the ribbons of a supertile go one level up. The combinatorial core of the proof, with these computations, is also formalised in the Lean proof assistant. The proof, the programs, the formalisation and this text were produced with an AI system; the author has not checked every step, and no mathematician has reviewed it (see *Use of AI and verification status* in Section 1).

1. Introduction

A *Celtic knot* is a pattern of interlaced ribbons [2, 5]. How many separate strands a finite design of this kind contains has been studied, for instance, by Fisher and Mellor [8]. A *knot tile* is a polygon carrying ribbon segments that end on its edges, so that copies of the tile, placed edge to edge, link up into longer ribbons. We ask for a knot tile whose ribbons, in a tiling of the whole plane, form one endless strand. In a periodic tiling this cannot happen: if the decorated tiling is invariant under a non-zero translation, its ribbons never form a single strand (Remark 3.5). So a tile that only tiles non-periodically is the natural place to look.

In 2023 Smith, Myers, Kaplan and Goodman-Strauss described the *hat*, an aperiodic monotile [17], and then the *Spectre* [18]: a shape that tiles the plane, admits only non-periodic tilings, and admits only tilings in which all tiles have the same handedness, even when reflections are allowed [18, Thm. 2.2]. For background on tilings and substitution see [10, 1]. The Spectre is a curved-edge form of the polygon Tile(1, 1) (Fig. 1): tilings by Spectres correspond exactly to the tilings by Tile(1, 1) in which all tiles have the same handedness [18, §2].

This paper describes a knot tile on the Spectre and proves that it is *mono-stranded* in the following sense: no tiling by tiles of one handedness contains a closed strand (the Endless Strand Theorem), and in one explicit tiling all its pieces form a single strand (the Ariadne

Date: 12 October 2026.

2020 *Mathematics Subject Classification.* Primary 52C20; Secondary 05B45, 52C23, 57K10.

Key words and phrases. aperiodic monotile, Spectre, substitution tiling, Celtic knot, knot tile, computer-assisted proof.

This paper is licensed under CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/); the programs that accompany it are under the MIT licence.

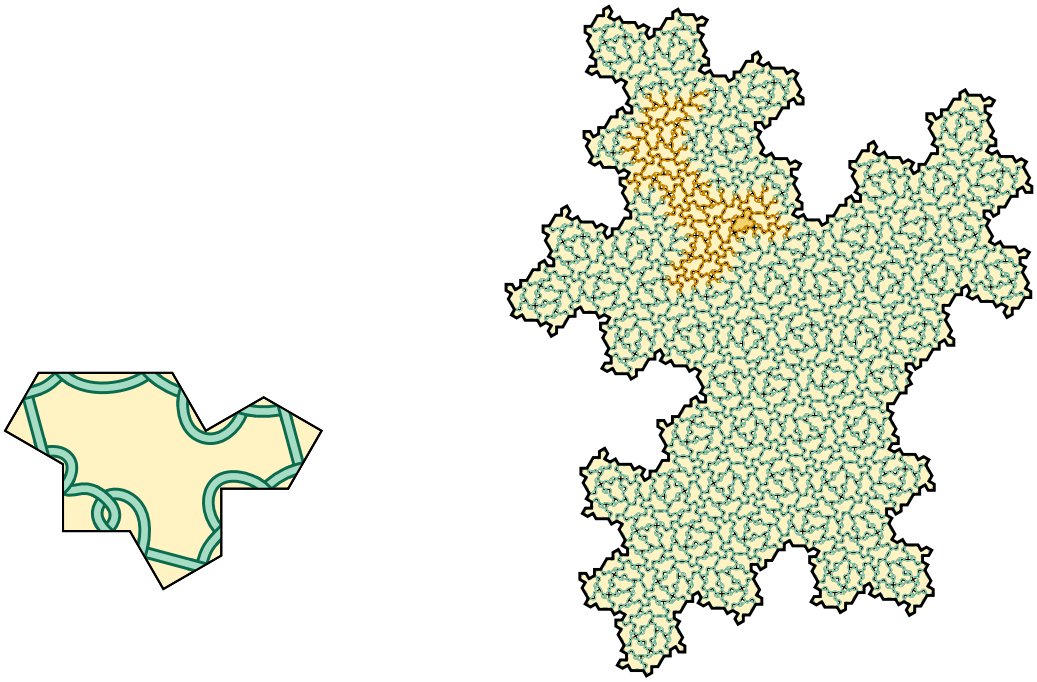


FIGURE 1. Left: the tile Tile(1, 1) with its fourteen ribbon pieces. Right: a supertile of 559 tiles (the block N_3 of Section 8). The original tile N_0 is shaded; all fourteen of its pieces already lie on a single arc (a path of ribbon from rim to rim) of this block, drawn in orange.

Theorem). We do not claim one strand for every tiling; computations indicate that some tilings have three (Remark 9.1). The decoration (Fig. 1) puts one crossing point on each of the fourteen edges and joins the 28 ribbon ends inside the tile by fourteen pieces: twelve *hooks*, each turning back round one vertex, and two pieces that cross each other inside the tile. Across an edge, the ribbons simply continue straight through the crossing point.

Our first result, the *Endless Strand Theorem*, concerns all such tilings. Every tiling by Tile(1, 1) with tiles of one handedness (a *homochiral* tiling; Section 2) groups uniquely into a hierarchy of larger and larger *supertiles* [18, Thm. 2.2]. Either any two tiles lie in a common supertile, or the tiling splits into two or more infinite supertiles, separated by a *fault* (Section 7.4); the Endless Strand Theorem covers both cases.

Endless Strand Theorem. *In every homochiral tiling of the plane by* Tile(1, 1)*, with every tile carrying the decoration of Fig. 1, there is no closed strand: every strand is a two-way infinite path of ribbon pieces.*

Our second result, the *Ariadne Theorem*, gives the single strand; it is named after Ariadne's thread, which led through the whole labyrinth. We prove it for one specific tiling, the *nest tiling* $\mathcal{N}$, which is grown from a single tile N_0 by repeatedly taking the Δ-supertile N_{b+1} that has N_b as its child number 1 (Section 8). In the usual language of substitution tilings, $\mathcal{N}$ is the tiling generated from one tile as a seed by iterating the substitution about it, a fixed point of the substitution (in the combinatorial sense; compare [1]); in terms of the supertile addresses of [19], it is the infinite-level supertile whose sequence of child numbers is $1, 1, 1, \ldots$. Figures 1, 2, 9 and 11 show blocks of it.

Ariadne Theorem. *In the nest tiling $\mathcal{N}$, with every tile carrying the decoration of Fig. 1, all the ribbon pieces of all the tiles form a single strand. It is a two-way infinite path that passes through every tile. More precisely, for every $b \geq 1$, all the arcs of the block N_b (the paths of ribbon through N_b from rim to rim; Section 4) lie on a single arc of N_{b+4}, and of N_{b+3} when b is even.*

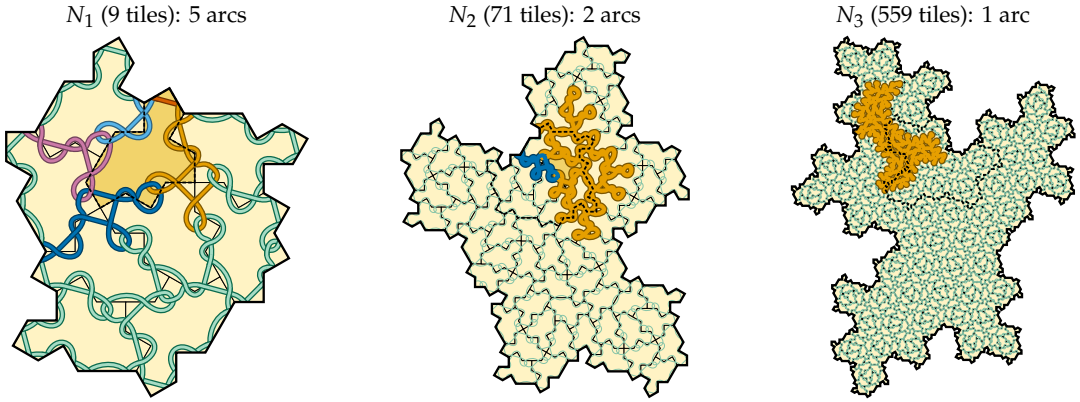


FIGURE 2. The arcs of the nest blocks N_1, N_2, N_3 that contain pieces of the original tile N_0 (shaded), one colour per arc. The dashed outline is the previous block.

Figure 2 shows the merging for N_0: its fourteen pieces lie on 5, 2 and 1 arcs of N_1, N_2 and N_3. Since the curves cross only transversally and in pairs, over and under can be assigned alternately, as in Celtic knotwork (Remark 3.6); the single strand then becomes an alternating diagram of a line.

The method. The proof has three ideas.

(1) *Black boxes.* Seen from outside, a supertile is just a list of which ribbon end on its boundary is joined to which. We show that every supertile of every kind and level has the same *signature*: a hook at every boundary vertex but two adjacent ones, A and B, where the two ribbons cross over; and no closed strand inside (Definition 4.3; the Signature Theorem, Theorem 7.6).
(2) *Zigzags.* When two such boxes are glued along a common boundary path, the ribbons zigzag along it, and the outcome depends only on the parity of its length (the Zigzag Lemma, Lemma 5.1). So lengthening a shared path by two edges changes nothing (Lemma 7.1).
(3) *Fixed parities.* The supertiles are combinatorially hexagons and compose by the substitution of [18, Fig. 5.1]. Their side lengths are given by Fibonacci numbers (Proposition 6.2); they only grow with the level, and their parities never change. So the side lengths at any level are reached from those at level 1 by lengthening, two edges at a time (Lemma 7.2), and a finite computation at level 1 gives the signature at every level.

For the Endless Strand Theorem one more idea is needed when the tiling has a fault: a closed strand that is short compared with the sides of large supertiles, and passes between them, would have to turn back somewhere; only a crossover makes a ribbon turn back (the Hairpin Lemma), and the crossovers of different supertiles are far apart (the Lonely Crossover Lemma; Section 7.4). The merging of all ribbons into one strand (Theorem 8.5) follows the same pattern. Where the ribbons of a block go one level up depends only on where they leave it, for most of its rim only on the side (Lemma 8.7); this map is computed at level 1 and carried to every level by the same lengthening. Followed three or four levels up, it puts every arc of a block on one strand.

Computer assistance. The proof rests on finitely many computer checks, all exact (with coordinates in $\mathbb{Z}[e^{i\pi/6}]$, or with integers): the signatures of the Mystic and of the level-1 supertiles, with their vertices, side lengths and crossover corners (Computation 7.5); and two computations on the 18 compositions of level-1 supertiles, built from the tables and the level-1 side lengths alone: their signature (Computation 7.3) and, for Δ, the junction map

(Computation 8.6). Each of these computations was done twice: the geometric ones by two separate programs, and the computations on compositions by a program and again in the kernel of the Lean proof assistant. The composition pattern is cited from [18]; two scripts that compare it with that paper's figures, on rendered images, are part of the proof. We also checked the conclusions directly on the actual supertiles up to level 9 (about 1.3×10^8 tiles), and by a second route, through skeletons and a symbolic computation of the merging, which the proof does not use (Remarks 7.8 and 8.9). Appendix A lists every computation with its script; the programs, with logs of a full run, are published with this paper at `https://cortado-j.github.io/SPECTRE_KNOT/` (folder `code/`, with the Lean formalisation in `lean/`).

Scope. The Endless Strand Theorem covers every homochiral tiling, including tilings with faults. The Ariadne Theorem is proved for the nest tiling only, and we make no claim about the number of strands in any other tiling. We do not expect the single strand in every tiling: computations indicate that in at least one other tiling, one without faults, the pieces form exactly three strands (Remark 9.1). See Section 9.

Related work. Curves drawn on tiles and joined across edges appear in several settings; each curve is then closed or two-way infinite, as in Lemma 3.4. For Truchet tilings the closed curves of random tilings have been studied by percolation and renormalisation methods [15, 11]; on rectangular grids the number of components of the mirror curves behind Celtic designs is computable, and without internal mirrors it is the greatest common divisor of the side lengths [12]. On the Penrose rhombs, the arc decoration of Conway and Penrose gives closed curves apart from at most two infinite ones (see [9] and the references there), and Girih interlace on a Penrose tiling produces many closed loops and knots [4]. Single curves through all the tiles of finite patches have been built as Hamiltonian cycles on Ammann–Beenker tilings [16] and as space-filling curves from decorations of substitution tilings [14]. Closest to the present work, [3] describes, in one nested tiling by Tile$(1, 1)$, a single non-crossing curve through every tile, using selected edges. Here every edge carries a ribbon end, the pieces cross, and the single strand is a knot diagram in the sense of Celtic knotwork. We are not aware of earlier full knotwork decorations that form a single strand.

Use of AI and verification status. The author came up with the original idea of looking for a mono-stranded aperiodic knot, and then used an AI system (Claude, by Anthropic) to search for it. The search produced 34 candidate designs; the author chose design 0.1:01 among them and drew its ribbons by hand in a tile editor. At the author's direction the AI system also wrote all the programs, developed the proof, wrote the Lean formalisation and drafted this paper.

The author has not checked every step of the proof, and parts of it are beyond the author's expertise. No mathematician has reviewed it, so its results should be regarded as claims, not as established theorems. The AI system reviewed its own argument in several passes. The programs and the logs of a full run are published with the paper. Where this paper calls a program or a check *independent* or *separate*, it means a separate implementation; all the programs were written by the same AI system.

Formal verification. The combinatorial part of the proof is formalised in the Lean 4 proof assistant [7], using its core library only (folder `lean/`, published with the programs). Lean's kernel has checked the following, in a combinatorial form in which a patch is described by which edge of which tile is glued to which edge of which other tile: every supertile of every kind and every level $m \geq 1$ has the signature (Theorem 7.6), and so contains no closed strand; in the union of any chain of supertiles, each a child of the next, in which every edge is eventually glued, every strand is a two-way infinite path (the Endless Strand Theorem for tilings whose hierarchy covers the plane); and in the nest tiling, built as the union of the blocks N_b, the pieces form one two-way infinite strand, with the bounds of

Theorem 8.5 (the Ariadne Theorem). The kernel also re-runs the finite computations that these use (Computations 7.3, 7.5 and 8.6). Not formalised: the link with tilings of the plane, that is, that the supertiles of an actual tiling are glued as Tables 1 to 3 say (this rests on [18] and Section 2, and is a stated assumption of the formal statements); the geometric Propositions 8.2 and 8.4 and part (a) of the Grandparent Lemma (Lemma 8.3); the Mystic (Lemma 4.4), which the formalisation does not treat separately; and the case of the Endless Strand Theorem for tilings with a fault (Lemmas 7.10 to 7.12). The proofs use only Lean's standard axioms. The formalisation was written by the AI system as well; Appendix A says how it relates to the text.

Outline. Section 2 recalls the facts about Spectre tilings that we use, in the notation of [18]. Section 3 defines the decoration and its strands. Section 4 introduces boxes and the signature. Section 5 proves the Zigzag Lemma and introduces skeletons, which the second routes and the formalisation use. Section 6 computes the side lengths. Section 7 proves the signature at every level, by lengthening, and the Endless Strand Theorem. Section 8 constructs the nest tiling and proves the Ariadne Theorem. Section 9 discusses scope and open questions. Appendix A lists every computation and the program that performs it.

2. Spectre tilings

2.1. **The tile.** Let $\rho = e^{i\pi/6}$. The polygon Tile(1, 1) has fourteen edges of unit length; we label its vertices $v_0, \dots, v_{13}$ counter-clockwise and its edges $e_i = v_i v_{i+1}$ (indices mod 14), with $v_0 = 0$ and

$$v_{i+1} - v_i = \rho^{k_i}, \qquad (k_0, \dots, k_{13}) = (0, 10, 1, 3, 0, 2, 5, 7, 4, 6, 6, 8, 11, 9).$$

All vertices lie in $\mathbb{Z}[\rho]$, and the edge directions are multiples of 30°; see Fig. 3. This is the equilateral member of the family Tile(a, b) [18, §§1–2], with its long side split into the collinear edges e_9, e_{10}, so that v_{10} has interior angle 180°; it is the mirror image of the polygon drawn in [18, Fig. 1.1], and has the handedness of the single Spectre and of the Mystic (a pair of tiles, defined below) drawn in [18, Fig. 2.1]. The clusters of [18, Figs. 2.1, 4.1] are drawn with tiles of the opposite handedness, so our level-1 supertiles (Section 2.2) are their mirror images; the handedness of the level-m supertiles alternates with m [18, Thm. 5.1]. *Spectres*, obtained by replacing every edge of Tile(1, 1) by a suitable curve, have tilings that correspond exactly, with the same placing isometries, to the *homochiral* tilings by Tile(1, 1), in which any two tiles are related by a rotation and a translation [18, §2, Lemma 2.1, p. 5]; so what [18] proves for tilings by Spectres holds for them, and we use it in that form. Reflecting the whole plane if necessary (a reflection maps strands to strands), we assume that every tile is the image of Tile(1, 1) under a rotation and a translation. Such tilings exist [18, Thm. 2.2], and in each the tiles meet full edge to full edge [18, proof of Lemma 2.1]. The tile has no non-trivial symmetry, proper or improper (its cyclic sequence of edge directions has none), so an isometry between two tiles is unique.

2.2. **Supertiles.** We use the hierarchy of [18]. There are nine *kinds*

$$\mathcal{K} = \{\Gamma, \Delta, \Theta, \Lambda, \Xi, \Pi, \Sigma, \Phi, \Psi\}.$$

A *level*-0 *supertile* of kind $K \neq \Gamma$ is a single tile; a level-0 supertile of kind Γ is a *Mystic* [18, Fig. 2.1], the pair of tiles $\{\text{Tile}(1,1),\ g(\text{Tile}(1,1))\}$ with $g(z) = \rho z + v_8$, and its congruent copies; we call Tile(1, 1) its *first* tile. (The Mystic has no non-trivial proper symmetry: one exchanging its tiles would be an involution taking Tile(1, 1) to g(Tile(1, 1)), but the only proper motion doing so is g, a rotation by 30°. So the first tile is determined by the pair.) The supertiles of a tiling are those of its hierarchy (Theorem 2.1(a) below): a level-$(m + 1)$ supertile is the union of level-m supertiles, its *children*, eight of them, or seven for kind Γ, of the kinds listed in Table 1, arranged as in Proposition 2.4. A level-1 supertile has the kind of its cluster in [18, Fig. 4.1], which is determined by the positions of its six vertices (defined below) [18, proof of Thm. 4.1]; a level-$(m + 1)$ supertile, $m \geq 1$, has kind T if its children

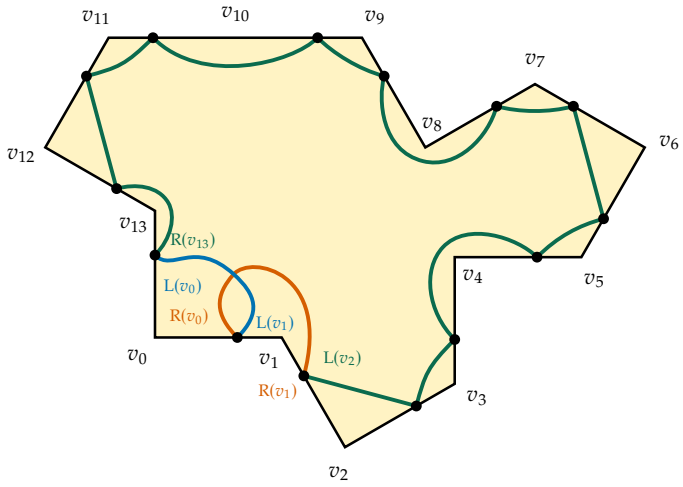


FIGURE 3. The decoration on the true geometry (centre lines of the ribbons of Fig. 1). Each edge carries one crossing point (dot), at which two pieces of the tile end: one leaves towards each end of the edge. On e_i these two ends are the ports $\mathrm{R}(v_i)$ (towards v_i) and $\mathrm{L}(v_{i+1})$ (towards v_{i+1}); the labels show them round the crossover corners. Green: the twelve hooks $\mathrm{L}(v_i) \leftrightarrow \mathrm{R}(v_i)$. Vermilion: $\mathrm{R}(v_0) \leftrightarrow \mathrm{R}(v_1)$; blue: $\mathrm{L}(v_0) \leftrightarrow \mathrm{L}(v_1)$, the crossover at the corners $A = v_0$ and $B = v_1$.

parent T	0	1	2	3	4	5	6	7
Γ	Π	Δ	Θ	Σ	Ξ	Φ	Γ	
Δ	Ξ	Δ	Ξ	Φ	Σ	Π	Φ	Γ
Θ	Ψ	Δ	Π	Φ	Σ	Π	Φ	Γ
Λ	Ψ	Δ	Ξ	Φ	Σ	Π	Φ	Γ
Ξ	Ψ	Δ	Π	Φ	Σ	Ψ	Φ	Γ
Π	Ψ	Δ	Ξ	Φ	Σ	Ψ	Φ	Γ
Σ	Ξ	Δ	Ξ	Φ	Σ	Π	Λ	Γ
Φ	Ψ	Δ	Ψ	Φ	Σ	Π	Φ	Γ
Ψ	Ψ	Δ	Ψ	Φ	Σ	Ψ	Φ	Γ
position in [18, Fig. 5.1]	left	Δ	bottom	right	Σ	top right	top left	Γ

TABLE 1. The children of each kind of supertile, and their places in the eight-hexagon layout of [18, Fig. 5.1] (Δ, Σ, Γ: the hexagons of those kinds). For $T = \Gamma$ there is no child at the bottom, and children $2, \ldots, 6$ are at the right, the Σ, the top right, the top left and the Γ.

form the panel T of [18, Fig. 5.1]. For $m \geq 1$, child c is the child at the position given in column c of the last row of Table 1; at level 1 the children are numbered as in Fig. 4. (This is the order used by Kaplan's implementation of the key-point construction of [18, App. A], see [13].) For definiteness a single tile gets the kind of its position in its level-1 supertile; only the distinction between Γ and the other kinds is used at level 0. In particular child 1 of every supertile is a Δ, and the last child is a Γ; at levels $m \geq 1$ a Δ occurs only as child 1.

The nine level-1 supertiles are the Spectre clusters (9 tiles) and the Mystic cluster (8 tiles, kind Γ) of [18, Fig. 2.1], with the markings of [18, Fig. 4.1], reflected (Section 2.1); they are drawn, with tiles of the handedness of Tile(1, 1), in Fig. 4. In the language of [18], our level-1

supertiles are the clusters, which are combinatorially the marked hexagons of [18, Fig. 4.2], and the level-n supertiles of hexagons of [18, §5] are our level-$(n+1)$ supertiles.

The following theorem restates, in our language, results of [18].

Theorem 2.1 (Smith, Myers, Kaplan, Goodman-Strauss [18])**.** *Let $\mathcal{T}$ be a homochiral tiling by* Tile$(1,1)$.

(a) [18, Thms. 2.2, 3.1, 4.1, §5, Fig. 2.1] *The tiles of $\mathcal{T}$ group uniquely into level-*1 *supertiles. Each contains one Mystic, in the position shown in* [18, Fig. 2.1]*; this Mystic and the other tiles, taken singly, are the level-*0 *supertiles inside it. For every $m \geq 1$ the level-m supertiles of $\mathcal{T}$ group uniquely into level-$(m+1)$ supertiles. Each tile lies in a unique nested sequence $X_0 \subset X_1 \subset \cdots$ with X_m a level-m supertile.*

(b) [18, Thms. 3.1, 4.1, 5.1, §5] *For every $m \geq 1$, the tiling of the plane by the level-m supertiles of $\mathcal{T}$ is combinatorially equivalent (homeomorphic as a topological complex* [18, §2.2]*) to a tiling by regular hexagons, each level-m supertile of kind K corresponding to a hexagon marked as the hexagon K of* [18, Fig. 4.2] *for even m, and as its mirror image for odd m, adjacent supertiles corresponding to hexagons sharing an edge with matching labels; and the grouping into level-$(m+1)$ supertiles corresponds, up to reflection, to the grouping of hexagons shown in* [18, Fig. 5.1].

In (a), [18, Thm. 4.1] gives the grouping into clusters; its uniqueness, like that of the higher levels, is part of [18, Thm. 2.2], and we use it only to speak of *the* hierarchy of $\mathcal{T}$. In (b), the handedness is that of our level-1 supertiles, and it alternates with m (Section 2.1).

In particular every level-m supertile ($m \geq 1$) is a closed topological disc whose boundary, its *rim*, is a simple closed polygon made of tile edges. Its *vertices* are the six points of the rim where it meets two other level-m supertiles, and its *sides* are the six paths along the rim between consecutive vertices. (The tile vertices on a rim are called *rim vertices*; for a supertile of level ≥ 1, "vertex" alone always means one of the six.) A vertex of one supertile that lies on the rim of another is a vertex of that one too, since exactly three supertiles meet there. Each side is shared, as a path of tile edges, with exactly one neighbouring supertile.

2.3. **Own orientation, side numbering and the crossover corner.** By [18, Thm. 5.1], the supertiles composed from a tiling by marked hexagons are combinatorially equivalent to the *mirror images* of those hexagons. So, at each level, the cyclic order of the labelled sides is reversed, and it is convenient to orient supertiles alternately.

Definition 2.2 (Own orientation)**.** The *own orientation* of a level-m supertile is counter-clockwise if m is even and clockwise if m is odd. We write $\omega_m = (-1)^m$.

Thus a child and its parent are always oriented oppositely. Each supertile has a distinguished rim vertex A, its *crossover corner*, defined recursively: for a tile, $A = v_0$; for a Mystic, A is the vertex v_0 of its first tile Tile$(1,1)$; for a level-$(m+1)$ supertile, A is the crossover corner of its Γ child. (That this point lies on the rim is computed for levels 0 and 1 and follows from Proposition 2.4(d) at higher levels.) The second corner B is the rim vertex next to A in the counter-clockwise direction; it appears in Theorem 7.6 (see Remark 7.7). We call A and B the *crossover corners*. The vertices of a level-m supertile ($m \geq 1$) are numbered $0, \ldots, 5$ in own order, starting with the first vertex at or after A; side j runs from vertex j to vertex $j+1$. Figure 4 shows this numbering for the level-1 supertiles.

2.4. **The composition pattern.** Theorem 2.1(b) says that every level of the hierarchy is assembled in the same combinatorial way. We record this in our numbering.

Definition 2.3 (Composition pattern)**.** The *composition pattern* C consists of the following data for each parent kind T, given in Tables 1 to 3 and drawn in Fig. 5.

(i) The kinds $K_0, \ldots, K_{n_T-1}$ of the children (Table 1).

(ii) For each child c and each side j of c, either the sibling c' and its side j' that lie across side j, or the symbol ∂ if side j lies on the parent's rim (Table 2).

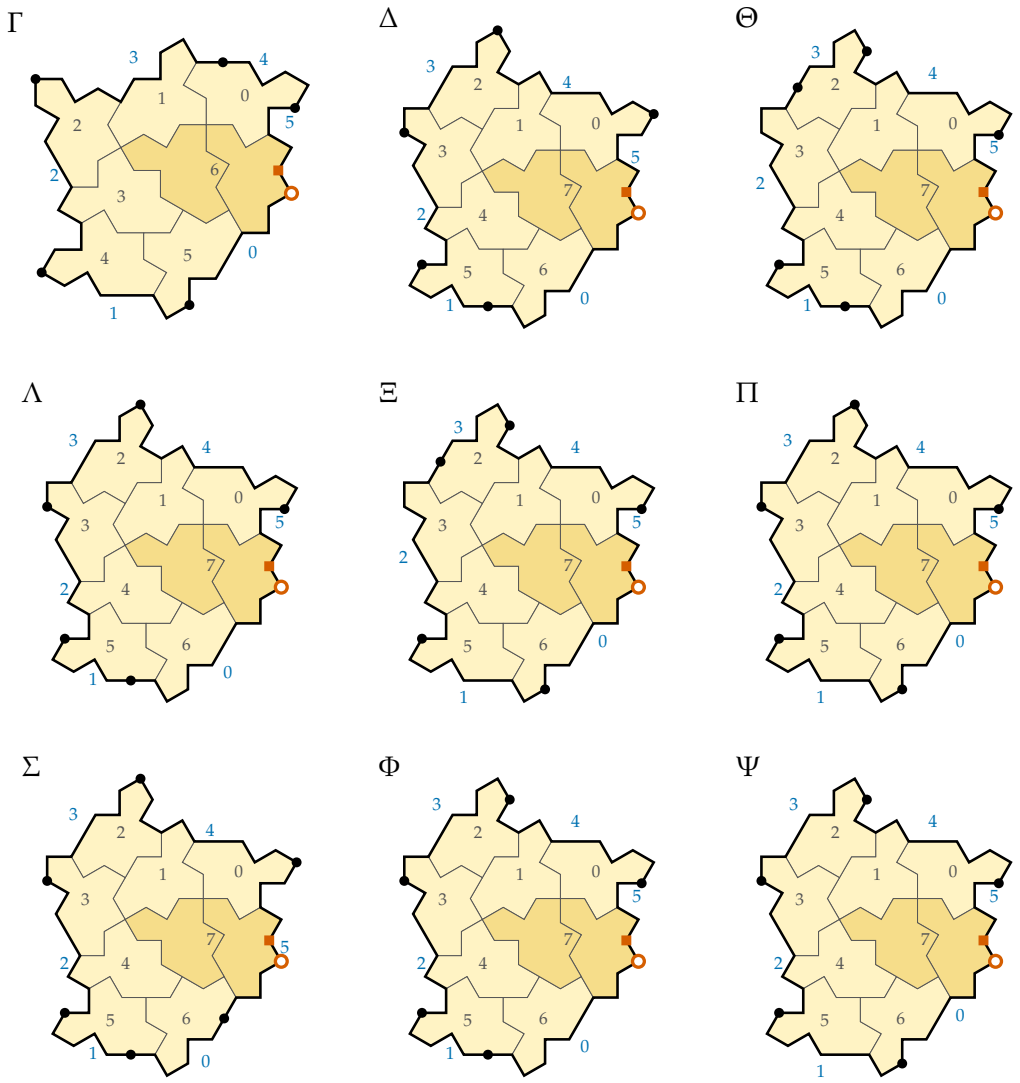


FIGURE 4. The nine level-1 supertiles, with true handedness. The Mystic is the darker pair of tiles. Grey numbers: the children (Table 1); black dots: the six vertices; blue numbers: the sides, numbered clockwise (the own orientation at level 1); open circle: the crossover corner A; filled square: the corner B (Theorem 7.6). For Σ, A lies inside side 5.

(iii) For each parent side J, the sequence of child sides (c, j) that make up side J, in the parent's own order (Table 3).

Proposition 2.4 (The pattern holds at every level)**.** *Let $\mathcal{T}$ be a homochiral tiling by* Tile(1, 1)*, let $m \geq 1$, and let Y be a level-$(m+1)$ supertile of $\mathcal{T}$ of kind T, with children $X_0, \ldots, X_{n_T-1}$. Then:*

(a) *X_c has kind K_c (Table 1);*
(b) *side j of X_c is the same path of tile edges as side j' of $X_{c'}$, traversed in the opposite direction, whenever Table 2 lists $c'.j'$, and it lies on the rim of Y when the table lists ∂;*
(c) *side J of Y is the concatenation of the children's sides listed in Table 3, and vertex J of Y is the vertex $j+1$ of the child c whose side $c.j$ is listed first;*
(d) *the crossover corner A of Y is vertex 0 of its Γ child, which is vertex 0 of Y if $T \neq \Sigma$, and lies inside side 5 of Y if $T = \Sigma$.*

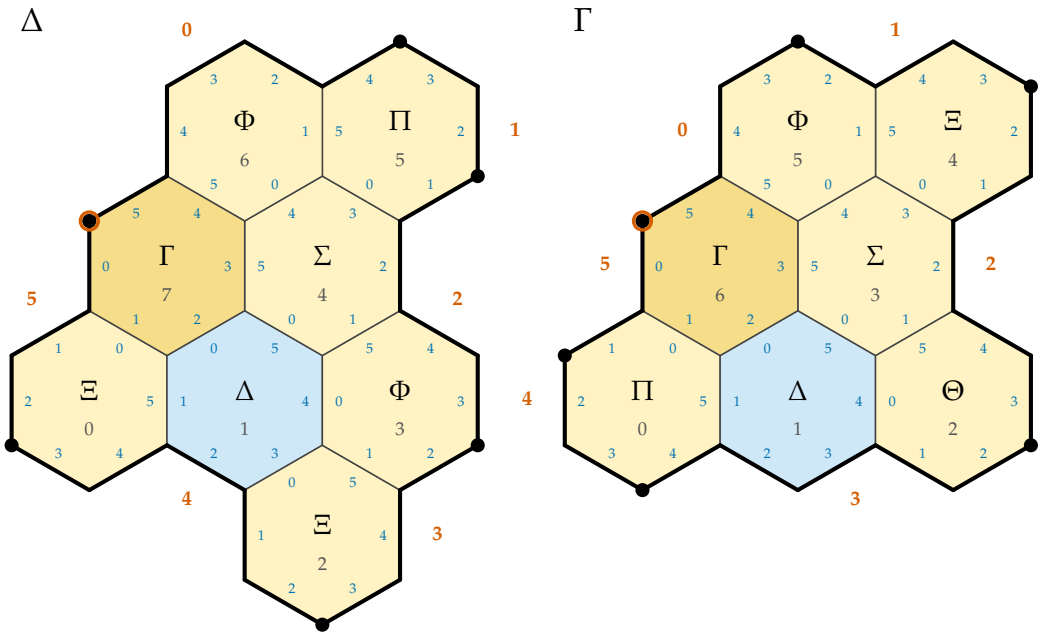


FIGURE 5. The composition pattern C for Δ (left; the same for all kinds except Γ, with the children's kinds as in Table 1) and for Γ (right), in the layout of [18, Fig. 5.1]. Each hexagon is a child: its kind, its number (grey), and its sides 0–5 (blue). Thick black: the parent's rim; black dots: the parent's vertices; vermilion numbers: the parent's sides; circled: the parent's crossover corner, which is vertex 0 of the Γ child. Children's sides are numbered in one rotational sense and the parent's in the other.

child	side 0	side 1	side 2	side 3	side 4	side 5
0	7.1	∂	∂	∂	∂	1.1
1	7.2	0.5	∂	2.0	3.0	4.0
2	1.3	∂	∂	∂	∂	3.1
3	1.4	2.5	∂	∂	∂	4.1
4	1.5	3.5	∂	5.0	6.0	7.3
5	4.3	∂	∂	∂	∂	6.1
6	4.4	5.5	∂	∂	∂	7.4
7	∂	0.0	1.0	4.5	6.5	∂

child	side 0	side 1	side 2	side 3	side 4	side 5
0	6.1	∂	∂	∂	∂	1.1
1	6.2	0.5	∂	∂	2.0	3.0
2	1.4	∂	∂	∂	∂	3.1
3	1.5	2.5	∂	4.0	5.0	6.3
4	3.3	∂	∂	∂	∂	5.1
5	3.4	4.5	∂	∂	∂	6.4
6	∂	0.0	1.0	3.5	5.5	∂

TABLE 2. Adjacency of the children's sides: the entry $c'.j'$ in row c, column j means that side j of child c is side j' of child c'; ∂ means that it lies on the parent's rim. Left: every parent kind except Γ. Right: Γ.

Justification. By Theorem 2.1(b), the level-m supertiles inside Y are combinatorially the hexagons of the corresponding panel of [18, Fig. 5.1]: hexagon edges correspond to sides

Eight children (every $T \neq \Gamma$): the parent's rim, in own order from vertex 0 of child 7.

position	0	1	2	3	4	5	6	7	8	9	10
segment	7.5	6.4	6.3	6.2	5.4	5.3	5.2	5.1	4.2	3.4	3.3
position	11	12	13	14	15	16	17	18	19	20	21
segment	3.2	2.4	2.3	2.2	2.1	1.2	0.4	0.3	0.2	0.1	7.0

Seven children ($T = \Gamma$), from vertex 0 of child 6; sides $0, \dots, 5$ begin at positions $0, 3, 6, 11, 16, 18$.

position	0	1	2	3	4	5	6	7	8	9
segment	6.5	5.4	5.3	5.2	4.4	4.3	4.2	4.1	3.2	2.4
position	10	11	12	13	14	15	16	17	18	19
segment	2.3	2.2	2.1	1.3	1.2	0.4	0.3	0.2	0.1	6.0

parent	first position of side					
	0	1	2	3	4	5
Δ	0	5	7	11	14	19
Θ	0	5	7	13	15	20
Λ	0	5	7	11	14	20
Ξ	0	3	7	13	15	20
Π	0	3	7	11	14	20
Σ	2	5	7	11	14	19
Φ	0	5	7	11	15	20
Ψ	0	3	7	11	15	20

TABLE 3. The parent's sides as concatenations of children's sides. A segment $c.j$ is side j of child c, which the parent's own orientation traverses from vertex $j+1$ to vertex j of the child. Side J of the parent consists of the segments from its first position up to the first position of side $J+1$ (cyclically). For $T \neq \Sigma$ the parent's side 0 begins at vertex 0 of the Γ child, which is the parent's crossover corner. For $T = \Sigma$ the crossover corner lies inside side 5 (positions 19–1), after the segments 0.2, 0.1, 7.0.

kind	side 0	side 1	side 2	side 3	side 4	side 5
Γ	α^-	α^+	γ^-	δ^-	β^+	β^-
Δ	γ^+	β^+	ϵ^-	α^+	γ^-	ζ^-
Θ	γ^+	β^+	θ^+	β^+	η	β^-
Λ	γ^+	β^+	ϵ^-	α^+	θ^-	β^-
Ξ	α^-	ϵ^+	θ^+	β^+	η	β^-
Π	α^-	ϵ^+	ϵ^-	α^+	θ^-	β^-
Σ	ζ^+	β^+	ϵ^-	α^+	γ^-	δ^+
Φ	γ^+	β^+	ϵ^-	ϵ^+	η	β^-
Ψ	α^-	ϵ^+	ϵ^-	ϵ^+	η	β^-

TABLE 4. The label of [18, Fig. 4.2] carried by side j of a supertile of each kind: the label of the edge of the hexagon in direction W, SW, SE, E, NE, NW for $j = 0, \dots, 5$.

and hexagon vertices to vertices. We use the following dictionary between the labels of [18, Fig. 4.2] and our side numbers:

(∗) side j of a level-m supertile of kind K carries the label of the edge of hexagon K in direction W, SW, SE, E, NE, NW for $j = 0, \dots, 5$ respectively (Table 4); its crossover corner A is vertex 0 if $K \neq \Sigma$; if $K = \Sigma$, A is not a vertex but lies in the relative interior of side 5.

We show (∗) by induction on m. For $m = 1$ it is a computation on the level-1 supertiles, which have 44 rim edges for Γ and 46 for the other kinds; their vertices lie at the rim positions, counted from A in own order,

$$\Gamma : 0, 7, 14, 24, 34, 39 \qquad \Delta : 0, 10, 15, 23, 30, 40 \qquad \Theta : 0, 10, 15, 26, 31, 41$$
$$\Lambda : 0, 10, 15, 23, 30, 41 \qquad \Xi : 0, 7, 15, 26, 31, 41 \qquad \Pi : 0, 7, 15, 23, 30, 41$$
$$\Sigma : 4, 10, 15, 23, 30, 40 \qquad \Phi : 0, 10, 15, 23, 31, 41 \qquad \Psi : 0, 7, 15, 23, 31, 41.$$

Drawn with the edges in odd directions lengthened to $\sqrt{3}$, which turns the tiles into hats and turtles [18, §3], and in the handedness of [18, Fig. 4.1], the nine level-1 supertiles coincide with the nine marked clusters there: the same outlines and the same tiles inside, the vertices at the marked dots, and the labels of Table 4 in the places where that figure has them; A is the dot between the W and NW sides (for Σ, a point of the NW side), on the light tile of the Mystic. A script checks this against a render of the figure (Appendix A). Suppose $(*)$ holds for the children of Y. The panel draws the inner hexagons with their kinds and edge markings but without their edge labels; the matching rules fix the rotation of each of them, and there is exactly one consistent choice (it is found by the script that encodes the panels, Remark 2.5). It is the same in all nine panels, and Fig. 5 shows the resulting side numbers, so that Tables 2 and 3 can be compared with [18, Fig. 5.1] by eye. With this dictionary, Tables 2 and 3 are then a transcription of the panel of [18, Fig. 5.1] for T, including its vertices (the six marked points on its outline); this gives (a)–(c). The crossover corner of Y is that of its Γ child, which is the child's vertex 0 by $(*)$, and Table 3 places this vertex at the start of the parent's side 0 if $T \neq \Sigma$ and inside side 5 if $T = \Sigma$; this is (d). Finally, the labels that the panel assigns to the parent's sides, numbered from this point in the parent's own order, are those of Table 4 for every T. So $(*)$ holds for Y. In particular A always lies on the rim. □

Remark 2.5 (Computer checks of the pattern). A script encodes all nine panels of [18, Fig. 5.1], finds the rotations of the inner hexagons used in the proof above, and compares the result with Tables 2 and 3; this is part of the proof. Since the tables carry the whole weight of the citation, we also checked them against supertiles of levels 1 to 8 built with the key-point construction of [18, App. A]. Both agree with the tables (Appendix A).

We shall use one consequence of Theorem 2.1(a) and Proposition 2.4 repeatedly.

Lemma 2.6 (Rigidity)**.** *Any two level-m supertiles of the same kind, in possibly different homochiral tilings, are related by a rotation and translation that maps descendants to descendants with their numbers (children, their children, and so on, down to the tiles). For $m \geq 1$ it maps the six vertices, the sides and the corner A of one onto those of the other.*

Proof. Induction on m. For $m = 0$ this is clear, since a Mystic is a congruent copy of a fixed pair of tiles with its first tile marked. For $m = 1$, the level-1 supertiles of a homochiral tiling are congruent copies of the clusters of Fig. 4 [18, Fig. 2.1, §4], with their vertices at the points marked in [18, Fig. 4.1]; the numbers of their children are those of the reference clusters of Fig. 4, and A is fixed by the Mystic. Let $m \geq 1$. By induction, the children of two level-$(m+1)$ supertiles Y, Y' of kind T are pairwise related by proper motions g_c, which map the vertices of child c of Y onto those of child c of Y' with their numbers. If side j of child c is glued to side j' of child c', then by Proposition 2.4(b), applied to Y and to Y', vertices j and $j+1$ of c coincide with vertices $j'+1$ and j' of c', in Y and in Y'. So g_c and $g_{c'}$ agree on two distinct points; being proper motions, they are equal. The children's adjacency graph is connected, so all the g_c are one motion g. By Proposition 2.4(c),(d), the vertices and the corner A of Y are vertices and the corner A of children, so g maps them, and hence the sides, onto those of Y'. Since g agrees with each g_c, it maps descendants to descendants with their numbers. □

Supertiles built by computer. The computations on tiles (the base case of $(*)$ and the child numbers at level 1, Proposition 6.1 and Lemma 6.4 at level 1, and Computation 7.5) use the level-1 supertiles built with the key-point construction of [18, App. A]. Their tiles, outlines and vertices are those of the clusters of [18, Figs. 2.1, 4.1] (proof of Proposition 2.4), so, by [18, Thm. 4.1], they are congruent, with their tiles, to the level-1 supertiles of every homochiral tiling. (Our programs find the level-1 vertices as the points where three level-1 supertiles meet inside built supertiles of levels 2 to 4; what certifies them is the comparison with [18, Fig. 4.1].) Higher levels enter the proofs only through the composition pattern.

Write $E_m(K)$ for the number of rim edges (equally, rim vertices) of a level-m supertile of kind K, and $\ell_m(K, j)$ for the number of edges of its side j. By Lemma 2.6 these numbers are well defined.

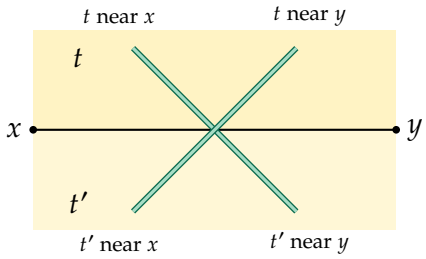


FIGURE 6. The straight-through rule at the crossing point of an edge xy: the end of t that leaves the crossing point towards x continues as the end of t' heading towards y.

3. The decoration and its strands

3.1. **Ports and pieces.** Let t be a tile and e one of its edges, with endpoints x and y. The tile has two *ports* on e, the port *near* x and the port *near* y; formally, they are the pairs (e, x) and (e, y). In the drawing (Fig. 3), e carries one crossing point, and the two ribbon ends of t that meet it, one running off towards x and one towards y, are these two ports. For a vertex v_i of the tile, write

$$\mathrm{R}(v_i) = \text{the port on } e_i \text{ near } v_i, \qquad \mathrm{L}(v_i) = \text{the port on } e_{i-1} \text{ near } v_i,$$

the ports just after and just before the corner v_i in counter-clockwise order. Each of the 28 ports is $\mathrm{R}(v_i)$ or $\mathrm{L}(v_i)$ for exactly one i.

Definition 3.1 (The decoration)**.** The *decoration* joins the 28 ports of Tile$(1,1)$ in pairs by fourteen *pieces*:

$$\underbrace{\mathrm{L}(v_i) \leftrightarrow \mathrm{R}(v_i) \quad (2 \le i \le 13)}_{\text{twelve hooks}}, \qquad \underbrace{\mathrm{R}(v_0) \leftrightarrow \mathrm{R}(v_1), \qquad \mathrm{L}(v_0) \leftrightarrow \mathrm{L}(v_1)}_{\text{the crossover}}.$$

Every tile of a homochiral tiling carries the image of this decoration.

In our drawing the two crossover pieces cross exactly once, transversally, and this is the only crossing inside the tile (Remark 3.6). Each hook turns back round one vertex. Figure 3 shows the pieces and their ports, and Fig. 1 the ribbons as drawn. The drawing uses cubic Bézier curves, with the crossing point of edge e_i at distance $0.5 + 0.15(-1)^i$ from v_i. Only the pairing of Definition 3.1 matters for the theorems; any drawing that realises it may be used.

3.2. **The straight-through rule.** Let $\mathcal{T}$ be a homochiral tiling, and let e be an edge shared by tiles t and t', with endpoints x and y.

Definition 3.2 (Straight-through rule)**.** The port of t on e near x is *joined* to the port of t' on e near y, and the port of t near y to the port of t' near x.

This is what a ribbon does when it continues straight through the crossing point (Fig. 6). In the drawing the two tiles' crossing points on e coincide: the tiles traverse e in opposite directions, and an edge e_i of one tile is always glued to an edge e_j of the other with $i + j$ odd. Indeed, by the vertex colouring of [18, proof of Lemma 2.1], the interior angles of Tile$(1,1)$ at even and odd vertices are multiples of $90°$ and of $120°$ respectively, and these classes agree where vertices meet; as v_i of one tile meets v_{j+1} of the other, i and $j+1$ have the same parity. The proofs use only Definitions 3.1 and 3.2.

3.3. **Strands.**

Definition 3.3 (Strand graph)**.** Let $\mathcal{T}$ be a homochiral tiling. The *strand graph* $\mathcal{G}(\mathcal{T})$ is the multigraph with one vertex for each piece of each tile and one edge for each pair of ports joined by the straight-through rule, between the two pieces having these ports as ends. A *strand* is a connected component of $\mathcal{G}(\mathcal{T})$; a strand is *closed* if it is finite.

Lemma 3.4. *Every vertex of $\mathcal{G}(\mathcal{T})$ has degree* 2*. Hence every strand is either a finite cycle (a closed strand) or a two-way infinite path.*

Proof. A piece has two ends, each a port of its tile on some edge e. In a tiling of the plane, e is shared by exactly one other tile, so each end is joined to exactly one port, an end of a piece of that neighbour. Joins always link pieces of different tiles, so $\mathcal{G}(\mathcal{T})$ has no loops, and every vertex has degree exactly 2. A connected, locally finite multigraph with all degrees 2 is a finite cycle (possibly of length 2) or a two-way infinite path. □

Remark 3.5 (A single strand needs a non-periodic tiling). Let $\mathcal{T}'$ be any edge-to-edge tiling of the plane by compact decorated tiles, each carrying at least one piece, with one crossing point on each edge and strands defined by the straight-through rule, and suppose that $\mathcal{T}'$, decoration included, is invariant under a translation $h \neq 0$. If its strand graph were a single two-way infinite path, with vertices $x \in \mathbb{Z}$, then h would act on it freely, since it moves every tile, and not as a reflection $x \mapsto c - x$, since h^2 fixes no piece; so h would act as $x \mapsto x + k$ with $k \neq 0$, and the pieces would fall into $|k|$ orbits. But the tiles fall into infinitely many orbits, since the cylinder $\mathbb{R}^2/\langle h \rangle$ has infinite area, and every tile has a piece. So a decorated tiling with a translational symmetry never has a single strand. Every tiling by Tile$(1, 1)$ of one handedness is non-periodic [18, Thm. 2.2], so this obstruction does not arise there.

Remark 3.6 (Over and under). In the drawing of Fig. 1 each piece stays inside its tile away from its ends, the two crossover pieces cross once, transversally, no other two pieces of a tile meet except at common ends, and at each crossing point the pieces meet the edge at ±45° (checked numerically; Appendix A). So the curves cross only transversally, in pairs, and the usual chequerboard rule makes over and under alternate along every strand [6]: the single strand of the Ariadne Theorem becomes an alternating diagram of a line, as in Celtic knotwork. None of this affects which pieces lie on which strand.

4. Boxes and the signature

4.1. **Patches and boxes.** A *patch* is a finite set P of tiles of a homochiral tiling whose union is a closed disc with a simple polygonal boundary, the *rim* ∂P. Every supertile of level ≥ 1 is a patch (Theorem 2.1), and so are a tile and a Mystic (whose two tiles share four consecutive edges, $e_6, \ldots, e_9$ of the first tile; its rim has 20 edges). Every rim edge belongs to exactly one tile of P. Let $E = E(P)$ be the number of rim edges, and number the rim vertices $u_0, \ldots, u_{E-1}$ in a chosen orientation of the rim, indices mod E. For a supertile we use its own orientation (Definition 2.2), except in Section 5, where a parent takes its children's orientation. For a rim vertex u_k define the *rim ports*

$$\mathrm{R}(u_k) = \text{the port on } u_k u_{k+1} \text{ near } u_k, \qquad \mathrm{L}(u_k) = \text{the port on } u_{k-1} u_k \text{ near } u_k,$$

where the port belongs to the tile of P containing that edge. There are $2E$ rim ports. For the tile itself, with its counter-clockwise orientation, these are the ports $\mathrm{R}(v_i), \mathrm{L}(v_i)$ of Section 3. Reversing the orientation exchanges R and L at every vertex.

Lemma 4.1 (The box of a patch)**.** *Start at a rim port of P and follow the pieces of the tiles of P, crossing every interior edge of P by the straight-through rule. The path ends at a different rim port. This defines an involution π_P on the rim ports without fixed points. The pieces of P that are not reached from any rim port form finitely many cycles, the* closed strands *of P; write $\kappa(P)$ for their number.*

Proof. The pieces of P and the joins across interior edges form a finite graph in which every piece has degree 2, except that a piece end at a rim port has no partner. Its components are therefore paths, whose two ends are rim ports, and cycles. Paths have two distinct ends. □

We call the paths of Lemma 4.1 the *arcs* of P; there are exactly E of them. The triple $(\partial P, \pi_P, \kappa(P))$ is the *box* of P. Everything we need about the inside of P is in its box, as the next lemma shows.

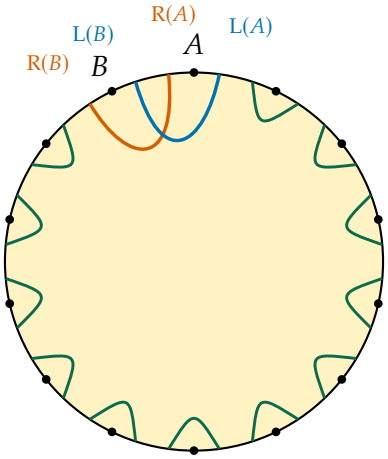


FIGURE 7. The signature at (A, B), drawn for a rim with 14 vertices oriented counter-clockwise, as for the tile ($A = v_0$, $B = v_1$): a hook at every rim vertex except A and B, and the two crossover arcs $\mathrm{R}(A) \leftrightarrow \mathrm{R}(B)$ and $\mathrm{L}(A) \leftrightarrow \mathrm{L}(B)$.

Lemma 4.2 (Gluing)**.** *Let a patch P be the union of patches $P_1, \dots, P_n$ with pairwise disjoint interiors. Then the box of P is determined by the boxes of the P_i together with the identification of their shared rim edges. More precisely, π_P is computed by alternately applying the π_{P_i} and the straight-through rule across shared edges, until a rim port of P is reached. The closed strands of P are those of the P_i together with the cycles of this alternation that contain no rim port of P.*

Proof. Every arc or closed strand of P is a concatenation of arcs and closed strands of the P_i, joined across edges shared by two of the P_i. □

4.2. **The signature.**

Definition 4.3 (Signature)**.** Let A and B be adjacent rim vertices of a patch P. We say that P has the *signature at* (A, B) if

(i) $\pi_P(\mathrm{L}(u)) = \mathrm{R}(u)$ for every rim vertex $u \notin \{A, B\}$ (a *hook* at u);
(ii) $\pi_P(\mathrm{R}(A)) = \mathrm{R}(B)$ and $\pi_P(\mathrm{L}(A)) = \mathrm{L}(B)$ (the *crossover*);
(iii) $\kappa(P) = 0$.

The definition is symmetric in A and B and does not depend on the orientation of the rim, since reversing it exchanges L and R. A patch with the signature has $E - 2$ hook arcs, one at each rim vertex other than A and B, and two crossover arcs. Figure 7 is a picture.

Lemma 4.4. *A tile has the signature at (v_0, v_1), and a Mystic has the signature at the vertices v_0, v_1 of its first tile.*

Proof. For the tile this is Definition 3.1. For the Mystic it is a finite computation with Lemma 4.2: there is no closed strand, the 18 rim vertices other than these two are hooks, and the crossover persists (Appendix A). □

5. The Zigzag Lemma and the skeleton

This section studies how the ribbons of glued boxes join. Section 5.1 proves the Zigzag Lemma, on which the proofs in Sections 7 and 8 rest. Section 5.2 condenses a composition to its *skeleton* and follows ribbons on it (Proposition 5.4). The proofs in this paper do not need the skeleton; the Lean formalisation uses Proposition 5.4, and the second routes of Remarks 7.8 and 8.9 use the skeleton.

Throughout this section a patch P is the union of patches (*children*) $X_0, \dots, X_{n-1}$ with disjoint interiors, all of them oriented in the same rotational sense, and P is oriented in that sense too. Each child X_c has the signature at (A_c, B_c). When P is a supertile and the

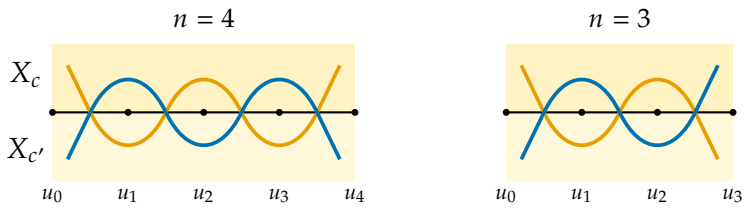


FIGURE 8. The Zigzag Lemma. Along a shared path whose interior corners are hooks on both sides, the ribbons cross the path at every edge and turn back at every corner, alternately in X_c (above) and $X_{c'}$ (below). For even n each connection returns to the side it started from; for odd n it ends on the other side.

X_c are its children, the children carry their own orientation, so P is oriented opposite to its own orientation. This does not matter for the signature, which does not depend on the orientation (Section 4); but it does exchange L and R at the parent's rim vertices.

5.1. **Runs and the Zigzag Lemma.** Two children $c \neq c'$ oriented in the same sense traverse a common rim path in opposite directions. For a rim vertex u of X_c we write $\mathrm{R}_c(u), \mathrm{L}_c(u)$ for the rim ports of X_c in the common orientation fixed above (for supertiles, their own orientation).

Lemma 5.1 (Zigzag Lemma)**.** *Let $u_0u_1 \cdots u_n$ ($n \geq 1$) be a path of n edges lying on the rims of both X_c and $X_{c'}$, traversed by X_c from u_0 to u_n. Suppose that none of the interior vertices $u_1, \ldots, u_{n-1}$ is a crossover corner of X_c or of $X_{c'}$. Then, alternately applying the straight-through rule across the edges u_ku_{k+1} and the hooks of X_c and $X_{c'}$ at $u_1, \ldots, u_{n-1}$, the ports at the two ends of the path are joined as follows:*

$$\begin{aligned} &n \text{ even:} && \mathrm{R}_c(u_0) \leftrightarrow \mathrm{L}_c(u_n), && \mathrm{R}_{c'}(u_n) \leftrightarrow \mathrm{L}_{c'}(u_0); \\ &n \text{ odd:} && \mathrm{R}_c(u_0) \leftrightarrow \mathrm{R}_{c'}(u_n), && \mathrm{L}_c(u_n) \leftrightarrow \mathrm{L}_{c'}(u_0). \end{aligned}$$

These two connections use every port on the edges of the path, so no closed strand lies along the path. Moreover, for $0 < k < n$, the hook of X_c at u_k lies on the connection through $\mathrm{R}_c(u_0)$ if k is even. If k is odd, it lies on the connection through $\mathrm{L}_c(u_n)$ when n is odd, and on the connection through $\mathrm{R}_{c'}(u_n)$ when n is even.

Proof. See Fig. 8. Start at $\mathrm{R}_c(u_0)$, the port of X_c on u_0u_1 near u_0. The straight-through rule takes it to the port of $X_{c'}$ on u_0u_1 near u_1. In the orientation of $X_{c'}$, which runs from u_1 to u_0 along this edge, that port is $\mathrm{R}_{c'}(u_1)$. If $n = 1$ we have reached $\mathrm{R}_{c'}(u_n)$. Otherwise u_1 is a hook of $X_{c'}$, which leads to $\mathrm{L}_{c'}(u_1)$, the port of $X_{c'}$ on u_1u_2 near u_1. Crossing u_1u_2 leads to the port of X_c on u_1u_2 near u_2, which is $\mathrm{L}_c(u_2)$. If $n = 2$ we stop. Otherwise the hook of X_c at u_2 leads to $\mathrm{R}_c(u_2)$, and the pattern repeats with all indices shifted by two. So the path visits the hooks of $X_{c'}$ at odd k and the hooks of X_c at even k. It ends at $\mathrm{L}_c(u_n)$ if n is even and at $\mathrm{R}_{c'}(u_n)$ if n is odd. The same argument from the other end, starting at $\mathrm{R}_{c'}(u_n)$ for even n and at $\mathrm{L}_c(u_n)$ for odd n, gives the second connection; it visits the remaining hooks. Each connection crosses each of the n edges once, using two of its four ports, so together they use all of them. □

5.2. **Marks, runs and the skeleton.**

Definition 5.2 (Marks and runs)**.** A *mark* is a point that is a rim vertex of some child and is either

(i) a *junction*: a rim vertex of a child X_c at which the two rim edges of X_c have different neighbours, where a neighbour is a sibling or "outside P"; or
(ii) one of the crossover corners A_c, B_c of some child.

A *run* of X_c is a path along the rim of X_c between two consecutive marks lying on that rim. (Each child has at least two marks on its rim, A_c and B_c, so runs are paths with distinct ends.)

Along a run the neighbour of X_c does not change. So a run either lies on the rim of P (a *rim run*) or is shared with exactly one sibling $X_{c'}$ (an *interior run*). In the second case the same path is also a run of $X_{c'}$, traversed in the opposite direction, because marks are points and the same marks bound it on both sides; we call it the *partner* of the run. At an interior vertex u of an interior run, X_c and $X_{c'}$ share both edges at u, so the full angle at u is filled by these two children and no third child touches u. Since crossover corners are marks, the Zigzag Lemma applies to every interior run.

Definition 5.3 (Skeleton)**.** The *skeleton* $\mathcal{S}$ of the composition $P = \bigcup X_c$ consists of the following data: the children and their kinds; for each child, the cyclic sequence of its marks and runs, and which of its marks are A_c and B_c; for each interior run, its partner run and the parity of its length; the cyclic order of the rim runs round ∂P.

Two skeletons of compositions with the same children (by number and kind) are *equal* if there is a bijection of their marks that preserves, for each child, the cyclic order of its marks and the labels A_c, B_c, and preserves partner runs, the parities of interior runs and the cyclic order of the rim runs. The maps τ and σ and the parent ports below are defined from these data alone, so such a bijection carries them, and hence the conclusions of Proposition 5.4, from one composition to the other.

The *reduced ports* are the ports $\mathrm{R}_c(u)$ and $\mathrm{L}_c(u)$ for every child c and every mark u on its rim. The port $\mathrm{R}_c(u)$ lies on the run of X_c that starts at u, and $\mathrm{L}_c(u)$ on the run that ends at u. Define two involutions on reduced ports.

- *Inside a child*: $\tau(\mathrm{L}_c(u)) = \mathrm{R}_c(u)$ if $u \notin \{A_c, B_c\}$; $\tau(\mathrm{R}_c(A_c)) = \mathrm{R}_c(B_c)$ and $\tau(\mathrm{L}_c(A_c)) = \mathrm{L}_c(B_c)$.
- *Across an interior run* from u to u' of X_c, of length n, with partner in $X_{c'}$: if n is even, $\sigma(\mathrm{R}_c(u)) = \mathrm{L}_c(u')$ and $\sigma(\mathrm{R}_{c'}(u')) = \mathrm{L}_{c'}(u)$; if n is odd, $\sigma(\mathrm{R}_c(u)) = \mathrm{R}_{c'}(u')$ and $\sigma(\mathrm{L}_c(u')) = \mathrm{L}_{c'}(u)$.

Both τ and σ are determined by $\mathcal{S}$. The *parent ports* are the reduced ports on rim runs. They are exactly the rim ports $\mathrm{R}(u), \mathrm{L}(u)$ of P at the rim vertices u of P that are marks (the *parent marks*), since X_c and P induce the same orientation on a rim run.

Proposition 5.4 (Reduced composition)**.** *Let $P = \bigcup X_c$, with every child having the signature. Then:*

(a) *every rim vertex of P that is not a mark is a hook of P;*
(b) *for every reduced port p, the ribbon of P followed from p (first through its child) reaches the rim of P at the first parent port among $\tau(p)$, $\tau\sigma\tau(p)$, $\tau\sigma\tau\sigma\tau(p), \dots$, if there is one (the ports $\sigma\tau(p), \dots$ in between lie on interior runs); for a parent port p this is $\pi_P(p)$;*
(c) *$\kappa(P)$ is the number of cycles of the alternation of σ and τ that contain no parent port.*

In particular, π_P at the parent marks and $\kappa(P)$ are determined by the skeleton.

Proof. (a) Let u be a rim vertex of P that is not a mark. The rim edge of P after u lies in some X_c; if the other rim edge of X_c at u were shared with a sibling, u would be a junction. So both rim edges of P at u belong to X_c, and u lies inside a rim run of X_c and is not a crossover corner of X_c. Hence $\pi_{X_c}(\mathrm{L}_c(u)) = \mathrm{R}_c(u)$, and both ports are rim ports of P. (b) and (c) follow from Lemma 4.2: the ribbon starting at p runs through X_c to a reduced port $\tau(p)$, since by the signature of X_c the arc of X_c starting at a port at a mark ends at a port at a mark. If $\tau(p)$ lies on an interior run, the ribbon then follows that run, by the Zigzag Lemma, to $\sigma\tau(p)$. A closed strand of P not inside a child must cross some shared edge, and by the Zigzag Lemma it then passes through a reduced port; following it from there gives a cycle of the alternation. Conversely each such cycle, read as arcs of children joined by zigzag connections, is a closed strand of P, and distinct cycles give distinct strands. The children have no closed strands. □

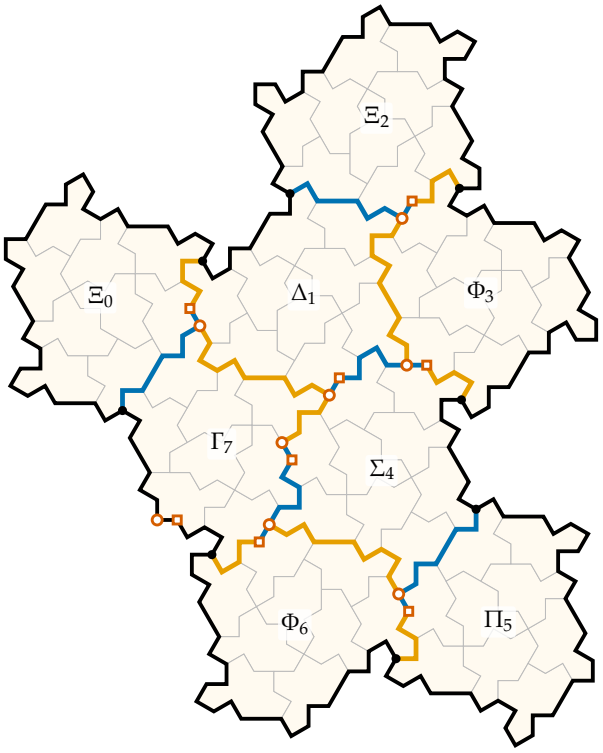


FIGURE 9. Marks and runs for the composition of a level-2 Δ-supertile from its eight level-1 children (subscripts are child numbers). Dots: the 24 marks (open circles and squares: the children's crossover corners A_c and B_c). Shared runs are coloured by the parity of their length: blue odd, orange even. Black: rim runs. The skeleton records this picture up to the lengths of the runs, keeping only the parities of the shared runs.

Corollary 5.5 (Where a hook goes)**.** *In the situation of Proposition 5.4, let u be a rim vertex of X_c that is not a mark. If u lies inside a rim run, the hook of X_c at u is a hook of P. If u lies inside an interior run, traversed by X_c from u_0 to u_n, at distance k from u_0, then the hook of X_c at u lies on the same arc or closed strand of P as the reduced port $\mathrm{R}_c(u_0)$ if k is even; if k is odd, as $\mathrm{L}_c(u_n)$ when n is odd and as $\mathrm{R}_{c'}(u_n)$ when n is even.*

Proof. This is the last part of Lemma 5.1, together with Proposition 5.4(a). □

6. Side lengths

6.1. **Letters.** Every side of a level-m supertile ($m \geq 1$) carries one of the edge labels of [18, Fig. 4.2] (Table 4). Ignoring the signs, there are eight *letters*, $\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \eta, \theta$. Two sides that are glued to each other carry labels x^+ and x^- (or η and η), and so the same letter; this is the matching rule of [18, §4] (Theorem 2.1(b)). For siblings it can also be read off Tables 1, 2 and 4; for supertiles with different parents, in particular across a fault (Section 7.4), we use it as cited.

Proposition 6.1 (Lengths by letter)**.** *For every $m \geq 1$ the length $\ell_m(K, j)$ depends only on the letter of side j of K. Writing x_m for the common length of the sides with letter x,*

$$
\begin{aligned}
\alpha_{m+1} &= \beta_m + \epsilon_m + \eta_m, & \beta_{m+1} &= \alpha_m + \epsilon_m,\\
\gamma_{m+1} &= \beta_m + 2\epsilon_m + \eta_m + \theta_m, & \delta_{m+1} &= \alpha_m + \beta_m + \epsilon_m + 2\theta_m,\\
\epsilon_{m+1} &= 3\epsilon_m + \eta_m, & \zeta_{m+1} &= \alpha_m + \epsilon_m + \theta_m,\\
\eta_{m+1} &= 4\epsilon_m + \eta_m, & \theta_{m+1} &= 4\epsilon_m + \eta_m + \theta_m,
\end{aligned}
\tag{1}
$$

with $(\alpha_1, \beta_1, \gamma_1, \delta_1, \epsilon_1, \zeta_1, \eta_1, \theta_1) = (7, 5, 10, 10, 8, 6, 10, 11)$.

Proof. For $m = 1$ this is a direct count on the level-1 supertiles (Fig. 4; Appendix A). By Proposition 2.4(c), $\ell_{m+1}(T, J)$ is the sum of the lengths of the children's sides listed in Table 3. Inspection of Tables 1, 3 and 4 shows that the multiset of letters of these children's sides depends only on the letter of side J of T, and is the one in (1). For example, side 0 of Δ (letter γ) is made of 7.5, 6.4, 6.3, 6.2, 5.4, with letters $\beta, \eta, \epsilon, \epsilon, \theta$. The claim follows by induction on m. □

The recurrence (1) has a surprisingly simple solution. Let F_n be the Fibonacci numbers, $F_1 = F_2 = 1$.

Proposition 6.2 (Fibonacci side lengths)**.** *For every* $m \geq 1$,

$$\alpha_m = F_{3m+2} + 2, \qquad \beta_m = F_{3m+1} + 2, \qquad \gamma_m = \delta_m = \eta_m = 2F_{3m+2},$$
$$\epsilon_m = F_{3m+3}, \qquad \zeta_m = 2F_{3m+1}, \qquad \theta_m = F_{3m+4} - 2.$$

Proof. For $m = 1$, $(F_4, F_5, F_6, F_7) = (3, 5, 8, 13)$ gives the initial values. For the induction step put $a = F_{3m+1}$ and $b = F_{3m+2}$, so that $F_{3m+3} = a + b$, $F_{3m+4} = a + 2b$, $F_{3m+5} = 2a + 3b$, $F_{3m+6} = 3a + 5b$ and $F_{3m+7} = 5a + 8b$. Substituting into (1):

$$\alpha_{m+1} = (a+2) + (a+b) + 2b = F_{3m+5} + 2, \qquad \beta_{m+1} = (b+2) + (a+b) = F_{3m+4} + 2,$$
$$\gamma_{m+1} = (a+2) + 2(a+b) + 2b + (a+2b-2) = 4a + 6b = 2F_{3m+5},$$
$$\delta_{m+1} = (b+2) + (a+2) + (a+b) + 2(a+2b-2) = 4a + 6b,$$
$$\eta_{m+1} = 4(a+b) + 2b = 4a + 6b, \qquad \epsilon_{m+1} = 3(a+b) + 2b = F_{3m+6},$$
$$\zeta_{m+1} = (b+2) + (a+b) + (a+2b-2) = 2F_{3m+4},$$
$$\theta_{m+1} = 4(a+b) + 2b + (a+2b-2) = 5a + 8b - 2 = F_{3m+7} - 2.$$ □

Since F_n is even exactly when $3 \mid n$:

Corollary 6.3 (Parities are constant)**.** *For every* $m \geq 1$ *the lengths* α_m, β_m, θ_m *are odd and* γ_m, δ_m, ϵ_m, ζ_m, η_m *are even.*

For example, the rim of a level-m Δ-supertile has length $E_m(\Delta) = \alpha_m + \beta_m + 2\gamma_m + \epsilon_m + \zeta_m$, which is 46, 182, 758, 3,198, 13,534, ...

6.2. **The crossover corner.**

Lemma 6.4 (Position of A)**.** *Let* $m \geq 1$. *For* $K \neq \Sigma$ *the crossover corner* A *of a level-*m K*-supertile is a vertex, hence its vertex* 0. *For* $K = \Sigma$ *it lies on side* 5 *(letter* δ*), at distance*

$$d_m = 2F_{3m+1}$$

from vertex 5 *and at distance* $\delta_m - d_m = 2F_{3m}$ *from vertex* 0.

Proof. For $K \neq \Sigma$ this is Proposition 2.4(d) for $m \geq 2$, and is read off Fig. 4 for $m = 1$. For $m = 1$, $d_1 = 6$ is read off the level-1 Σ-supertile. By Proposition 2.4(d) and Table 3, the crossover corner of a level-$(m+1)$ Σ-supertile follows the children's sides 0.2, 0.1, 7.0 on its side 5. Child 0 is a Ξ and child 7 a Γ, so these sides have letters θ, ϵ, α and $d_{m+1} = \theta_m + \epsilon_m + \alpha_m = (a+2b-2) + (a+b) + (b+2) = 2F_{3m+4}$, in the notation of the previous proof. □

Lemma 6.5 (Lower bounds)**.** *For every* $m \geq 1$*: the shortest side length at level* m *is* $\beta_m \geq 5$*;* $d_m \geq 6$*; and* $\delta_m - d_m \geq 4$. *Moreover* $d_m \geq \beta_m$*, and for* $m \geq 2$ *also* $\delta_m - d_m \geq \beta_m$.

Proof. By Proposition 6.2, $\alpha_m - \beta_m = F_{3m}$, $\zeta_m - \beta_m = F_{3m+1} - 2$ and $\epsilon_m - \beta_m = F_{3m+2} - 2$ are positive, and $\gamma_m = \delta_m = \eta_m = 2F_{3m+2}$ and $\theta_m = F_{3m+4} - 2$ exceed $\beta_m = F_{3m+1} + 2$; so β_m is the shortest, and $\beta_m \geq \beta_1 = 5$. Also $d_m \geq d_1 = 6$ and $\delta_m - d_m = 2F_{3m} \geq 2F_3 = 4$. Finally $d_m - \beta_m = F_{3m+1} - 2 > 0$, and for $m \geq 2$ $\delta_m - d_m - \beta_m = 2F_{3m} - F_{3m+1} - 2 = F_{3m-2} - 2 > 0$. □

7. The signature at every level, and no closed strands

This section proves the Signature Theorem and, from it, the Endless Strand Theorem. Section 7.1 describes how a supertile is glued from its children purely in terms of the side lengths ℓ_m and the offset d_m of Section 6. Section 7.2 shows, with the Zigzag Lemma, that lengthening a side by two edges preserves the signature of such a composition, and that the lengths at every level are reached from those at level 1 by such steps; so one finite computation at level 1 suffices. Section 7.3 runs the induction on the level. Section 7.4 deals with closed strands that pass between supertiles, which can occur only in a tiling with a fault.

7.1. **Compositions as a function of side lengths.** Fix a parent kind T, with children $c = 0, \dots, n_T - 1$ of kinds K_c (Table 1); exactly one of them, c_Σ, is a Σ. Let L assign a positive integer $L(c, j)$ to each child c and side $j \in \{0, \dots, 5\}$, so that sides glued by Table 2 have the same length; let d be an integer with $2 \le d \le L(c_\Sigma, 5) - 2$; and let $\omega \in \{\pm 1\}$. The *composition* $C_T(L, d, \omega)$ is the following abstract patch of boxes. Child c has a rim of $E_c = \sum_j L(c, j)$ vertices, numbered $0, \dots, E_c - 1$ in its own order; side j runs from vertex $V_c(j) = \sum_{i<j} L(c, i)$ to vertex $V_c(j+1)$. Its crossover corner A_c is vertex 0, or for $c = c_\Sigma$ the vertex at distance d from vertex 5 along side 5, and $B_c = A_c + \omega$. Each child is a box with the signature at (A_c, B_c) (Definition 4.3). Sides are glued in pairs as in Table 2, reversing direction: the o-th edge of side j of c is the o-th edge from the end of side j' of c'. The sides marked ∂ form the rim of the composition, in the order of Table 3; a rim vertex of the composition is the point between two consecutive rim edges, and its ports R and L are the ports, on these two edges, of the children they belong to. Ribbons are followed as in Lemma 4.2: alternately through a child, by its signature, and straight across a glued edge, until the rim is reached. This defines the box of the composition: the pairing of its rim ports and its closed strands. Lemma 5.1, Proposition 5.4, and Corollary 5.5 and their proofs apply unchanged to compositions (with marks, runs and skeletons defined as in Definitions 5.2 and 5.3), since they use only the signatures of the children and the straight-through rule. We say that $C_T(L, d, \omega)$ *has the signature* if it has no closed strand, and every rim vertex is a hook except the crossover corners A_Γ, B_Γ of its Γ child, which are adjacent rim vertices (B_Γ following A_Γ in the children's order if $\omega = +1$, and preceding it if $\omega = -1$) and carry the crossover.

Write ℓ_m also for the function $(c, j) \mapsto \ell_m(K_c, j)$. By Propositions 2.4 and 6.1 and Lemma 6.4, if the children of a level-$(m+1)$ T-supertile have the signature at $(A_c, A_c + \omega_m)$ in their own orientation, then the supertile is glued from its children as $C_T(\ell_m, d_m, \omega_m)$ is (with all children in their own orientation). Indeed: (i) by Proposition 2.4(a),(b) and Proposition 6.1 the children have the kinds and side lengths of the composition, and side j of X_c is side j' of $X_{c'}$ traversed oppositely, so points are matched as above; (ii) the rims of two children meet only in the closed sides they share: at a point inside a side both rim edges are shared with the partner, so the two children fill a neighbourhood, and a vertex lies on exactly three level-m supertiles, any two of which share a side ending there (Theorem 2.1(b)); (iii) so each side of a child is wholly shared with one sibling or wholly on the rim, as in Table 2; (iv) A_c is placed by Lemma 6.4, and $B_c = A_c + \omega_m$ by hypothesis; (v) the rim of the parent is given by Table 3 (Proposition 2.4(c)). Hence the children's rims, their corners and the gluing of their sides are those of $C_T(\ell_m, d_m, \omega_m)$. By Lemma 4.2, the strands of the supertile, followed from child to child, are then those of the composition; in particular the box of the supertile is the box of $C_T(\ell_m, d_m, \omega_m)$.

7.2. **Lengthening.** Let (L, d) be as above. A *move* lengthens one side of one child by two edges, together with the side glued to it if there is one. An edge e of that side is replaced by a path of three edges through two new vertices; e must not be the first or the last edge of its side (in each child that contains it), nor one of the two edges at A_Σ (the crossover corner of the Σ child). The result is (L', d'), where L' is L with the side (and its partner) two edges longer, and $d' = d + 2$ if e lies on side 5 of the Σ child before A_Σ, and $d' = d$ otherwise.

Lemma 7.1 (Lengthening)**.** *If $C_T(L, d, \omega)$ has the signature, then so does $C_T(L', d', \omega)$, for every move $(L, d) \to (L', d')$.*

Proof. Write $C = C_T(L, d, \omega)$ and $C' = C_T(L', d', \omega)$, and let u_0u_1 be the edge e. In C' it is replaced by the path $u_0v_1v_2u_1$; positions after e on the rim of each child containing it move on by two, and every vertex of C has an image vertex in C'. The corners of C' are the images of those of C: for A_Σ this is the choice of d', and $B_c = A_c \pm 1$ stays next to A_c, because the edge between them (the first edge of side 0, the last of side 5, or an edge at A_Σ) is never cut. In particular v_1 and v_2 are not crossover corners. Let φ map each port of C to the port of C' of the same hand at the image vertex. Then φ commutes with the pairings inside the children, since the corners correspond, and with the straight-through rule across every glued edge other than e. If e is shared by children c and c', crossing it joins $\mathrm{R}_c(u_0)$ to $\mathrm{R}_{c'}(u_1)$ and $\mathrm{L}_c(u_1)$ to $\mathrm{L}_{c'}(u_0)$ (the Zigzag Lemma, Lemma 5.1, with $n = 1$); in C', by Lemma 5.1 with $n = 3$, whose interior vertices v_1, v_2 are hooks of both children, the ribbon from $\varphi(\mathrm{R}_c(u_0))$ runs along the new path to $\varphi(\mathrm{R}_{c'}(u_1))$, and the ribbon from $\varphi(\mathrm{L}_c(u_1))$ to $\varphi(\mathrm{L}_{c'}(u_0))$. If e lies on the rim, v_1 and v_2 are hooks of the child, and so new rim vertices of C' that are hooks of C'.

Hence, for every port x of C, the ribbon of C' from $\varphi(x)$ passes through the images of the ports met from x in C, and reaches the rim at the image of the rim port reached in C. So the pairing of the rim ports of C' at the images of the rim vertices of C is the image of that of C: a hook at every one of them except A_Γ and B_Γ, which are adjacent as before and carry the crossover; and the new rim vertices, if any, are hooks. Finally, C' has no closed strand. Every port of C' at an old vertex is some $\varphi(x)$, whose ribbon reaches the rim. A strand through a port at v_1 or v_2 either is a hook of the rim (if e lies on the rim), or follows one of the two connections of Lemma 5.1 along $u_0v_1v_2u_1$, which use every port on these edges and end at ports at u_0 and u_1; so it too reaches the rim. □

Call (L, d) *admissible* if sides glued by Table 2 have the same length; $L(c, j) \geq \ell_1(c, j)$ and $L(c, j) \equiv \ell_1(c, j) \pmod 2$ for all c and j; $d \geq d_1$ and $d \equiv d_1 \pmod 2$; and $L(c_\Sigma, 5) - d \geq \ell_1(c_\Sigma, 5) - d_1 = 4$. Then $d \geq 6$ and $L(c_\Sigma, 5) - d \geq 4$, so the composition $C_T(L, d, \omega)$ is defined.

Lemma 7.2 (Reachability)**.** (a) *For every $m \geq 1$, the pair (ℓ_m, d_m) is admissible.*

(b) *Every admissible $(L, d) \neq (\ell_1, d_1)$ arises by a move from an admissible pair (L_0, d_0) with $\sum_{c,j} L_0(c, j) < \sum_{c,j} L(c, j)$. The move can be chosen so that its edge e lies at least two edges from both ends of its side, in each child that contains it, unless that side is side 5 of the Σ child or is glued to it.*

Hence every admissible pair is reached from (ℓ_1, d_1) by finitely many such moves through admissible pairs.

Proof. (a) Glued sides have the same letter (Section 6), so the same length. By Proposition 6.2 the letter lengths do not decrease with m, and by Corollary 6.3 their parities do not change. By Lemma 6.4, $d_m = 2F_{3m+1} \geq 6 = d_1$ is even, and $\ell_m(c_\Sigma, 5) - d_m = \delta_m - d_m = 2F_{3m} \geq 4$.

(b) If $L = \ell_1$, then $L(c_\Sigma, 5) = \delta_1 = 10$ and $d_1 \leq d \leq 10 - 4$, so $d = d_1$. Hence $L \neq \ell_1$, and by the parity condition some side j of some child c has $L(c, j) \geq \ell_1(c, j) + 2$. We undo a move on that side. If the side is neither side 5 of Σ nor glued to it, take $d_0 = d$ and the move whose edge is the third edge of the side (the third from the end in the partner); before the move the side has length at least $\ell_1(c, j) \geq 5$ (Lemma 6.5), so this edge lies at least two edges from both ends, in either child. Otherwise we undo a move on side 5 of the Σ child (and the side glued to it, which belongs to another child), measuring positions along side 5 from its vertex 5. If $d \geq d_1 + 2$, take $d_0 = d - 2$ and the move whose edge is the second edge of the side, before A_Σ; it does not touch A_Σ, since $d_0 \geq 6$. If $d = d_1$, take $d_0 = d$ and the second edge after A_Σ; it is not the last edge of the side, since $L(c_\Sigma, 5) - d \geq \ell_1(c_\Sigma, 5) + 2 - d_1 = 6$. In each case (L_0, d_0) is admissible, and $\sum L_0 = \sum L - 2$ or $\sum L - 4$. □

Computation 7.3 (The compositions at the level-1 lengths)**.** For each of the nine kinds T and both signs ω, the composition $C_T(\ell_1, d_1, \omega)$ has the signature.

This was computed twice: by a program that builds the 18 compositions from Tables 1 to 3 and the lengths alone, follows the ribbon from every one of their ports and checks each condition of the signature (Appendix A), and in the kernel of the Lean proof assistant (Section 1, *Formal verification*). Each composition has 8 children (7 for Γ), 732 ports (640) and a rim of 182 edges (160). As a negative control, with each child's corner B_c placed two vertices from A_c instead of one, all 18 checks fail.

Proposition 7.4 (Compositions at every level)**.** *For every $m \geq 1$, every kind T and both signs ω, the composition $C_T(\ell_m, d_m, \omega)$ has the signature.*

Proof. By Computation 7.3 and Lemma 7.1, every composition reached from (ℓ_1, d_1) by moves has the signature. By Lemma 7.2 this includes $C_T(\ell_m, d_m, \omega)$. □

7.3. **The induction.**

Computation 7.5 (Base case)**.** Every level-1 supertile has the signature at (A, B), where A is as in Lemma 6.4 and $B = A + \omega_1$ in own order. Moreover ℓ_1 and d_1 are as stated in Proposition 6.1 and Lemma 6.4.

This was computed from the tiles, using Lemma 4.2, in exact arithmetic; the program checks the statement as worded: the two non-hooks are the crossover corners of the Γ child (the Mystic), they are adjacent, $B = A + \omega_1$, they carry the crossover, and there is no closed strand. A second, separate program, in floating point, confirms it (Appendix A). The level-1 supertiles, as glued patches of tiles, are also the input of the Lean formalisation, whose kernel checks this computation again. The rims have $E_1 = 44$ edges for Γ and 46 for the other kinds.

Theorem 7.6 (Signature Theorem)**.** *In every homochiral tiling by* Tile$(1, 1)$*, every level-m supertile of every kind has the signature at (A, B), where A is its crossover corner (Section 2.3; placed as in Lemma 6.4 for $m \geq 1$) and $B = A + \omega_m$ in its own order. In particular no supertile contains a closed strand.*

Proof. For $m = 0$ this is Lemma 4.4, and for $m = 1$ it is Computation 7.5. Let $m \geq 1$, assume the theorem for level m, and let Y be a level-$(m+1)$ supertile of kind T. Its children have the signature at $(A_c, A_c + \omega_m)$, so by Section 7.1 its box is that of $C_T(\ell_m, d_m, \omega_m)$, which has the signature by Proposition 7.4. Hence Y has no closed strand, and hooks at all its rim vertices except the crossover corners A_Γ, B_Γ of its Γ child, which are adjacent and carry the crossover. By definition A_Γ is the crossover corner of Y, placed as in Lemma 6.4. Finally $B_\Gamma = A_\Gamma + \omega_m$ in the child's own order, which is opposite to that of Y; so $B = A - \omega_m = A + \omega_{m+1}$ in the own order of Y. □

Remark 7.7. Since the own orientation alternates with m and so does ω_m, in absolute terms B is always the counter-clockwise neighbour of A on the rim, as it is for the tile $(B = v_1)$.

Remark 7.8 (Other checks of the signature). Of the computations in this section the proof uses only Computations 7.3 and 7.5. The following computations and arguments agree with them, and are not used. (1) The compositions $C_T(\ell_m, d_m, \pm 1)$ were built and followed port by port for $m \leq 4$; each of the 1,792 single moves from (ℓ_1, d_1) was checked to keep the signature, with both signs ω; and the moves that the proof of Lemma 7.2 chooses were followed from (ℓ_2, d_2) and (ℓ_3, d_3) back to (ℓ_1, d_1) for every kind, each one checked to be a move and, from (ℓ_2, d_2), every composition on the way checked to have the signature. (2) The signature of the actual supertiles was computed from the tiles at levels 1 to 4 in exact arithmetic, and at levels 1 to 9 in floating point, by the two programs of Appendix A. (3) A second route to Proposition 7.4 uses the skeleton (Definition 5.3). For letter lengths λ (and $C_T(\lambda, d, \omega)$ the composition with $L(c, j) = \lambda$ of the letter of side j of K_c) with $\lambda(x) \geq 3$ and $3 \leq d \leq \lambda(\delta) - 3$, the marks on each side of a child lie at the symbolic positions $0 < 1 < d-1 < d < d+1 < \ell-1 < \ell$ or $0 < 1 < \ell-d-1 < \ell-d < \ell-d+1 < \ell-1 < \ell$, whose

order does not depend on the lengths, and the length of a run is a sum of side lengths plus a difference of symbolic positions. So the skeleton of $C_T(\lambda, d, \omega)$ depends only on λ mod 2, d mod 2 and ω, and by Proposition 5.4 so do its pairing at the parent marks and its closed strands (every other parent rim vertex is a hook). The 18 skeletons at $(\ell_1, d_1, \pm 1)$ have 24 marks (21 for Γ) and 52 runs (45), of which 10 (9) lie on the rim; their reduced walks give the signature, and they coincide with the skeletons of the actual supertiles at every level from 2 to 9 (a comparison that does not distinguish the parent's A from its B; the children's are distinguished).

7.4. **No closed strands.** We now prove the Endless Strand Theorem. By the Signature Theorem no closed strand lies inside one supertile. In most tilings this already suffices; a closed strand that passes between supertiles needs another argument.

Definition 7.9 (Faults)**.** The hierarchy of a homochiral tiling *covers the plane* if the supertiles containing some (equivalently, every) tile exhaust the plane. The union of the supertiles containing a given tile is an *infinite supertile*; if the hierarchy does not cover the plane, there are at least two, and the union of their boundaries is the *fault*.

(The two forms are equivalent, since every tile lies in some $X_m(t)$, as shown below, and two tiles that lie in a common supertile have the same supertiles from that level on; for the same reason two infinite supertiles are equal or have disjoint interiors.) Tilings with faults exist (Section 9).

If the hierarchy covers the plane, the Signature Theorem (Theorem 7.6) excludes closed strands at once. Indeed, let t be a tile whose supertiles $X_0(t) \subset X_1(t) \subset \cdots$ have union $\mathbb{R}^2$. Every tile t' lies in some $X_m(t)$: a point x of the interior of t' lies in $X_m(t)$ for some m, hence in some tile s of $X_m(t)$; every neighbourhood of x meets the interior of s, so the interiors of s and t' meet, and $s = t'$. A closed strand consists of finitely many pieces in finitely many tiles, so it lies in a single $X_m(t)$, and all its joins are across interior edges of $X_m(t)$; it is then a closed strand of $X_m(t)$, which Theorem 7.6 excludes. (This is the case of the Endless Strand Theorem that is formalised; Section 1, *Formal verification*.)

In a tiling with a fault, a closed strand need not lie in a single supertile, so the Signature Theorem alone does not exclude it. The argument below treats all tilings alike, and does not use the definitions above: a closed strand that is short compared with the sides of the level-m supertiles, and crosses between them, would have to turn back somewhere; only a crossover can make it turn back; and the crossovers of different supertiles are far apart. Recall that $\beta_m = F_{3m+1} + 2$ is the shortest side length at level m (Lemma 6.5).

The *rim graph* at level m is the union of the rims of all level-m supertiles.

Lemma 7.10 (Localisation Lemma)**.** *Let C be a closed strand of a homochiral tiling, with r pieces, and let $m \geq 1$ with $\beta_m > r$. Suppose that C does not lie in a single level-m supertile. Then the rim edges across which C passes from one level-m supertile to another form a tree G with at most r edges, which lies in one side or in the union of the three sides at one supertile vertex. So the tiles of C lie in two level-m supertiles that share a side, or in three around a common vertex.*

Proof. Follow C and cut it at the tile edges where it passes from one level-m supertile to another. There is at least one, so C is a cyclic sequence of arcs of level-m supertiles Y (Lemma 4.1), joined across *crossing edges*, each of which lies on the rims of two different supertiles. Each arc is a hook or a crossover arc of Y (Theorem 7.6), so the two crossing edges at its ends are rim edges of Y with a common vertex: for a hook at u they are the two rim edges at u, and for a crossover arc they are two of the three rim edges at A and $B = A \pm 1$. Hence the crossing edges form a connected graph G with at most r edges. It lies in the rim graph, which by Theorem 2.1(b) is a graph in which the vertices of the supertiles have degree 3, all other rim vertices degree 2, and any path between two different supertile vertices runs along a whole side, of at least $\beta_m > r$ edges. So G contains at most one supertile vertex u, and lies in one side, or in the union of the three sides at u. Every cycle of the rim

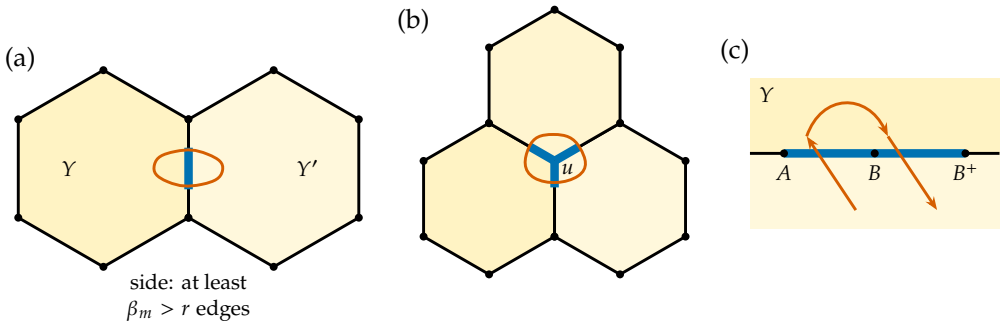


FIGURE 10. The Localisation and Hairpin Lemmas, with supertiles drawn schematically as hexagons. A closed strand with r pieces (vermilion) passes between supertiles only across a connected set of at most r rim edges (blue), while every side has more than r edges. So the crossings lie (a) inside one side, between the two supertiles that share it, or (b) near one vertex u, among the three supertiles around it. (c) A ribbon (vermilion) crosses the edge AB into Y towards A, the crossover arc of Y carries it back towards B, and it leaves across the next edge towards B^+: the walk of Lemma 7.11 runs B, A, B, B^+ and turns back at A (in the notation of the lemma, $X = A$, $Z = B$, $Z' = B^+$).

graph contains a supertile vertex, since the other rim vertices lie inside sides, which are paths; leaving that vertex along a side, the cycle runs to the far end of the side. So every cycle has more than r edges, and G is a tree. Every level-m supertile containing a tile of C is entered by C across a crossing edge on its rim. If G lies in one side, the only such supertiles are the two that share it. Otherwise they are among the three supertiles around u, each side at u being shared by two of them (Fig. 10(a),(b)). □

Lemma 7.11 (Hairpin Lemma)**.** *In the situation of Lemma 7.10, follow C and record a walk W in G: crossing an edge xy of G from the port near x to the port near y (the straight-through rule) is a step from x to y, and a crossover arc of a supertile Y, which joins a port near A_Y to a port near B_Y (Definition 4.3), is a step between A_Y and B_Y along the edge A_YB_Y. Then W is a closed walk of positive length in G, and:*

(a) W turns back, *that is, leaves a vertex along the edge by which it arrived, only at a crossover corner A_Y or B_Y of a supertile Y whose crossover arc C traverses;*
(b) *if C traverses a crossover arc of Y, then A_YB_Y is an edge of G, and W visits a vertex whose distance in the tree G from the midpoint of A_YB_Y is $\frac{3}{2}$.*

Proof. C crosses at least one edge, so W has positive length; its crossing steps run along edges of G, and so do its crossover steps, as shown below. Between two crossings C runs along one arc of a level-m supertile Y, a hook or a crossover arc (Theorem 7.6). A hook at a rim vertex u joins the ports near u on the two rim edges at u; so W arrives at u along one of these edges and leaves along the other, and does not turn back. Let $A = A_Y$ and $B = B_Y$. In absolute terms each crossover arc of Y joins the port of the edge AB near one of its ends, X say, to the port near the other end, Z, on the other rim edge ZZ' at Z: for $B = A + 1$ in the own order of Y, the arc R(A)–R(B) has $X = A$ and $Z = B$, and the arc L(A)–L(B) has $X = B$ and $Z = A$; for $B = A - 1$ the two arcs exchange these roles. When C traverses such an arc from X to Z, it arrives at the port near X on AB by crossing AB from the port near Z, and it leaves from the port near Z on ZZ' by crossing ZZ'; so W runs through Z, X, Z, Z', and in the other direction through Z', Z, X, Z. In both cases W turns back at X and not at Z, the edges AB and ZZ' lie in G, and W visits Z', whose distance in G from the midpoint of AB is $\frac{3}{2}$ (the path from Z' to the midpoint runs through Z, since G is a tree). These are all the places where consecutive steps of W meet, so (a) and (b) follow (Fig. 10(c)). □

Lemma 7.12 (Lonely Crossover Lemma)**.** *Let $m \geq 2$, and let $Y \neq Y'$ be level-m supertiles of a homochiral tiling, with crossover corners A_Y and $A_{Y'}$ (Section 2.3). Every path in the rim graph from A_Y to $A_{Y'}$ has at least β_m edges.*

Proof. Write $A = A_Y$ and $A' = A_{Y'}$. As in the proof of Lemma 7.10, the supertile vertices are the vertices of degree 3 of the rim graph, and a point inside a side lies on exactly two level-m supertiles, the two that share the side. Adjacent supertiles share whole sides with matching labels, that is, the same letter with opposite signs, or both η (Theorem 2.1(b), [18, Fig. 4.2]). By Table 4, side 0 of every kind other than Σ is labelled α^- or γ^+ and side 5 is labelled β^- or ζ^-, and side 5 of Σ is labelled δ^+; no two of these labels match.

Suppose first that neither Y nor Y' is a Σ. By Lemma 6.4, A and A' are vertex 0 of Y and of Y'. If $A = A'$, then Y and Y' are two of the three supertiles at this vertex, so they share a side ending there (Theorem 2.1(b)); it is side 0 or 5 of each, and the labels do not match. So $A \neq A'$, and a path from A to A', leaving the supertile vertex A along a side, runs along that whole side before it can reach another supertile vertex; it has at least β_m edges.

Suppose now that Y is a Σ; the case of Y' is symmetric. By Lemma 6.4, A lies inside side 5 of Y, at distances d_m and $\delta_m - d_m$ from its ends, and both are at least β_m (Lemma 6.5; here $m \geq 2$ is used). A path from A runs along side 5 of Y until it reaches A' or an end of that side. In the second case it has at least β_m edges. In the first, A' is an interior point of side 5 of Y, so it is not a supertile vertex: Y' is a Σ with A' inside its side 5, and Y' is the supertile across side 5 of Y. Then side 5 of Y is glued to side 5 of Y', and both are labelled δ^+, which do not match. □

Proof of the Endless Strand Theorem. Let C be a closed strand of a homochiral tiling, with r pieces, and choose $m \geq 2$ with $\beta_m > r$. If C lies in one level-m supertile, Theorem 7.6 excludes it. Otherwise let G and W be as in the Localisation Lemma and the Hairpin Lemma (Lemmas 7.10 and 7.11). For any point o of the tree G, at a vertex of W farthest from o both neighbours of that vertex along W are the neighbour on the path towards o, so W turns back there. Hence, by Lemma 7.11(a), C traverses a crossover arc of some supertile Y. If it also traversed one of a supertile $Y' \neq Y$, then A_Y and $A_{Y'}$ would lie in G (Lemma 7.11(b)), joined by a path in G of at most $r < \beta_m$ edges, contrary to the Lonely Crossover Lemma (Lemma 7.12). So every crossover arc that C traverses belongs to Y, and W turns back only at A_Y or B_Y. Take for o the midpoint of A_YB_Y: the vertex of W farthest from o is A_Y or B_Y, at distance $\frac{1}{2}$ from o. But by Lemma 7.11(b), W visits a vertex at distance $\frac{3}{2}$ from o, a contradiction. So there is no closed strand, and by Lemma 3.4 every strand is a two-way infinite path. □

8. The nest tiling and the single strand

This section builds the nest tiling (Section 8.1), shows that the arcs of every block merge within three or four levels (Section 8.2), and concludes the Ariadne Theorem (Section 8.3). The merging uses the device of Section 7: a map computed once at the level-1 lengths and carried to every level by the moves of Lemma 7.2.

8.1. **The nest.** Child 1 of every supertile is a Δ (Table 1).

Construction 8.1 (The nest)**.** Let $\mathcal{T}$ be a homochiral tiling [18, Thm. 2.2]. For each $h \geq 0$, let $Y^{(h)}$ be a level-h Δ-supertile of $\mathcal{T}$; one exists, for instance child 1 of any level-$(h+1)$ supertile. Let $Y^{(h)} = Y_h^{(h)} \supset Y_{h-1}^{(h)} \supset \cdots \supset Y_0^{(h)}$ be its chain of descendants, each child 1 of the next. Let g_h be the proper motion taking the tile $Y_0^{(h)}$ onto Tile$(1,1)$ (with v_0 at the origin), and put $N_b^{(h)} = g_h(Y_b^{(h)})$ for $b \leq h$.

Proposition 8.2. *$N_b^{(h)}$ does not depend on $h \geq b$ (or on the choices made). Writing N_b for it, $N_0 = \{\text{Tile}(1,1)\}$, and N_b is child 1 of N_{b+1} for every b.*

Proof. Let $h, h' \geq b$. By Lemma 2.6 there is a proper motion f with $f(Y_b^{(h)}) = Y_b^{(h')}$ that maps descendants to descendants with their numbers; hence $f(Y_{b'}^{(h)}) = Y_{b'}^{(h')}$ for all $b' \leq b$, in particular for the tiles Y_0. So $g_{h'}f$ and g_h both take the tile $Y_0^{(h)}$ onto Tile(1, 1), and since a tile has no symmetry they are equal. Thus $N_b^{(h')} = g_{h'}f(Y_b^{(h)}) = N_b^{(h)}$. □

We call N_b the *nest block* of level b, or simply a *block*. The *nest tiling* is $\mathcal{N} = \bigcup_{b\geq 0} N_b$, a set of tiles. Figure 11 shows three consecutive blocks.

Lemma 8.3 (Grandparent Lemma)**.** *Let $b \geq 1$.* (a) *The block N_b does not meet the rim of N_{b+2}.* (b) *Every edge of a tile of N_b is shared by two tiles of N_{b+2}.*

Proof. (a) By Table 2, the sides $0, 1, 3, 4, 5$ of child 1 face the siblings $7, 0, 2, 3, 4$, and side 2 lies on the parent's rim. A point of N_b that is not on its closed side 2 is therefore an interior point of N_{b+1}. This is clear for interior points of N_b and for interior points of the other sides, where N_b and one sibling together fill a neighbourhood. At the vertices $0, 1, 4, 5$ of N_b, exactly three level-b supertiles meet; two of them are the distinct siblings across the adjacent sides, and together they fill a neighbourhood. Hence

$$N_b \cap \partial N_{b+1} = \text{closed side 2 of } N_b. \tag{2}$$

By Table 3 (the Δ row: positions 14 to 18), side 4 of the parent is the concatenation of $2.2, 2.1, 1.2, 0.4, 0.3$. Since 1.2 is neither first nor last, the closed side 2 of N_b lies in the relative interior of side 4 of N_{b+1}. Now let $p \in N_b \cap \partial N_{b+2}$. Then $p \in \partial N_{b+1}$, for otherwise p would be interior to $N_{b+1} \subset N_{b+2}$. So p lies on side 4 of N_{b+1} by the above, and, by (2) for N_{b+1}, on the closed side 2 of N_{b+1}. But the closed sides 2 and 4 of N_{b+1} are disjoint: they are paths on a simple closed curve, separated by side 3 in one direction and by sides $5, 0, 1$ in the other.

(b) follows from (a). It also follows from the tables alone, and this is the form used in the formalisation: a rim edge of N_b lies on a side $j \neq 2$ of N_b, which faces a sibling inside N_{b+1}, or on side 2, which lies on side 4 of N_{b+1} (position 16 of Table 3); and side 4 of child 1 faces sibling 3 inside N_{b+2} (Table 2). □

Proposition 8.4. *$\mathcal{N}$ is a homochiral tiling of the plane, and any finitely many tiles of $\mathcal{N}$ lie in a common block N_b.*

Proof. The blocks are nested, so any finitely many tiles of $\mathcal{N}$ lie in a common block; in particular any two have disjoint interiors. We show that the tiles cover the plane. Fix h and work in the homochiral tiling $\mathcal{T}_h = g_h(\mathcal{T})$, which contains $N_0, \ldots, N_h$. Call two tiles of $\mathcal{T}_h$ *neighbours* if they share a point, and let dist be the resulting graph distance. We claim that every tile at distance at most k from N_1 lies in N_{1+2k}, provided $1 + 2k \leq h$. For $k = 0$ there is nothing to prove. If t has distance $k + 1$, it shares a point p with a tile of N_{1+2k}. By the Grandparent Lemma (Lemma 8.3(a)), p is an interior point of N_{3+2k}, so the interior of t meets the interior of a tile of N_{3+2k}; hence t is that tile. Now let $x \in \mathbb{R}^2$ and let D be the disc of radius $r = |x| + 1$ about the origin. The tiles of $\mathcal{T}_h$ meeting D have diameter $3 + \sqrt{3} < 5$ and area $3 + 3\sqrt{3} > 8$, so there are at most $k_0 = \lfloor \pi(r + 5)^2/8 \rfloor$ of them. They cover the connected set D, so their neighbour graph is connected. One of them is $N_0 \subset N_1$, since the origin is a vertex of N_0. So every tile of $\mathcal{T}_h$ meeting D, in particular one containing x, lies in N_{1+2k_0} as soon as $h \geq 1 + 2k_0$. Since N_{1+2k_0} does not depend on h, x is covered by $\mathcal{N}$. □

The blocks are supertiles: for $h \geq b$, $N_b = g_h(Y_b^{(h)})$ is a level-b Δ-supertile of the homochiral tiling $\mathcal{T}_h = g_h(\mathcal{T})$, and it is child 1 of N_{b+1} there. Since the arcs and closed strands of a patch depend only on its tiles, Proposition 2.4 and Theorem 7.6 apply to the blocks. In particular $\mathcal{N}$ has no closed strand: by the Endless Strand Theorem, since $\mathcal{N}$ is a homochiral tiling; or directly, since a closed strand passes through finitely many tiles, which lie in a common block N_b by Proposition 8.4, and would be a closed strand of N_b.

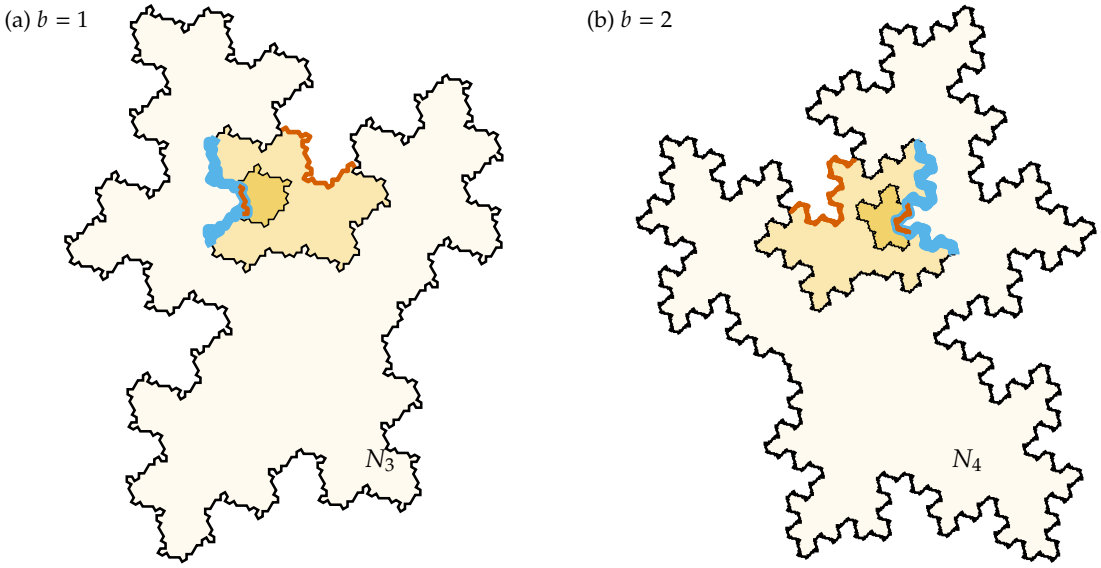


FIGURE 11. Three consecutive blocks of the nest: (a) $N_1 \subset N_2 \subset N_3$ and (b) $N_2 \subset N_3 \subset N_4$. Vermilion: side 2 of the two smaller blocks, which is the only side of each block on the rim of the next. Blue: side 4 of the middle block. Since the vermilion side of the smallest block lies inside the blue side, and the blue side faces a sibling inside the largest block, the smallest block does not reach the rim of the largest (Lemma 8.3).

b	tiles in N_b	arcs of N_b	$k = 1$	$k = 2$	$k = 3$	$k = 4$
0	1	14	5	2	1	1
1	9	46	10	2	2	1
2	71	182	37	2	1	1
3	559	758	146	2	2	1
4	4,401	3,198	613	2	1	1
5	34,649	13,534	2,586	2	2	1
6	272,791	57,318	10,949	2	1	–
even $b \geq 2$				2	1	1
odd $b \geq 1$				2	2	1

TABLE 5. The number of arcs of N_{b+k} on which the arcs of N_b lie, that is, the number of parts of $\mathcal{P}_k(b)$ (row $b = 0$: the 14 pieces of the single tile, computed on the drawing up to level 5). Rows $1 \leq b \leq 6$: computed by following every ribbon, up to level 9 (in floating point; for $b \leq 3$ also in exact arithmetic, up to level 7). Last two rows: the proof of Theorem 8.5, for every $b \geq 1$ of each parity.

8.2. **Merging.** For $b \geq 1$ and $k \geq 0$, every arc of N_b is contained in a unique arc of N_{b+k}, because N_{b+k} has no closed strand. Let $\mathcal{P}_k(b)$ be the partition of the $E_b(\Delta)$ arcs of N_b in which two arcs are in the same part when they lie on the same arc of N_{b+k}. The parts only merge as k increases.

Theorem 8.5 (Merging). *For every $b \geq 1$, $\mathcal{P}_4(b)$ has a single part; if b is even, so does $\mathcal{P}_3(b)$.*

Table 5 shows the numbers. The proof looks one level up at a time. In N_{m+1}, every arc of its child N_m lies on one of a few hooks of N_{m+1}, and which one depends only on where the

arc of N_m	m even ($\omega_m = +1$)	m odd ($\omega_m = -1$)
crossover arc through R(A) and R(B)	17	21
crossover arc through L(A) and L(B)	9	16
hook at u, $0 < u \le V_1$	17	21
hook at u, $V_1 < u \le V_2$	17	17
hook at u, $V_2 < u < V_3$	a hook of N_{m+1} itself	
hook at u, $V_3 \le u < V_4$	16	16
hook at u, $V_4 \le u < V_5$	12	16
hook at u, $V_5 \le u < E$	9	16

TABLE 6. The junction map (Lemma 8.7): the junction q such that an arc of N_m lies on the hook H_q of N_{m+1}. Hooks are listed by the position u of their rim vertex, which is never A or B.

arc leaves N_m: for most of the rim, only on the side of N_m. We compute this map once, at the level-1 lengths, and carry it to every level by the moves of Section 7.

Junctions. Let $m \ge 1$. The rim of N_{m+1}, in its own order from its corner A, is the concatenation of the 22 children's sides listed in the Δ row of Table 3, at positions $q = 0, \dots, 21$. The *junction* q is the rim vertex of N_{m+1} at which the side at position q starts (not to be confused with the junctions of Definition 5.2(i)). The junctions $9, 12, 16, 17, 21$ lie inside sides of N_{m+1} (Lemma 8.8 below), so they are hooks of N_{m+1} (Theorem 7.6); we write H_q for the hook arc of N_{m+1} at junction q. The side at position 16 is 1.2, side 2 of N_m, so junctions 16 and 17 are its two ends: vertices 3 and 2 of N_m. Positions on the rim of N_m are taken in its own order from its corner A, so that its vertices $0, \dots, 5$ lie at positions $0 = V_0 < V_1 < \cdots < V_5 < E$, where $E = E_m(\Delta)$; its corner $B = A + \omega_m$ is at position 1 if m is even and $E - 1$ if m is odd.

Computation 8.6 (The junction map at the level-1 lengths)**.** In $C_\Delta(\ell_1, d_1, \omega)$, for both signs ω, with child 1 in the place of N_m and positions and junctions defined in the same way: every rim vertex of child 1 strictly inside its side 2 is a hook of the composition; and the ribbon from every other port of child 1 reaches the rim of the composition at junction q, where q is given by Table 6 with $\omega_m = \omega$.

This was computed twice, like Computation 7.3: by the program of Appendix A, and in the kernel of the Lean proof assistant (its second part; the formalisation proves the first for every level, from the signature of child 1). As a negative control, exchanging the junctions 12 and 16 in Table 6 makes the check fail for both signs.

Lemma 8.7 (The junction map)**.** *Let $m \ge 1$. In N_{m+1}, every hook of N_m at a rim vertex strictly inside side 2 of N_m is a hook of N_{m+1}, and every other arc of N_m lies on the hook H_q of N_{m+1}, where q is given by Table 6.*

Proof. Say that (L, d) has the *junction property* if the statement of Computation 8.6 holds for $C_\Delta(L, d, \omega)$ for both ω. The children of N_{m+1} have the signature (Theorem 7.6), so by Section 7.1 N_{m+1} is glued from them as $C_\Delta(\ell_m, d_m, \omega_m)$ is, with N_m as child 1; and a ribbon of the composition from a port p of child 1 runs through child 1 along the arc of N_m ending at p, and on along the arc of N_{m+1} that contains it. A hook arc has both its ends at the same rim vertex. So the lemma says that (ℓ_m, d_m) has the junction property. By Computation 8.6, (ℓ_1, d_1) has it. By Lemma 7.2, (ℓ_m, d_m) is reached from (ℓ_1, d_1) by moves as in Lemma 7.2(b) through admissible pairs, so it suffices to show that such a move keeps the junction property. Child 1 is a Δ, and the side glued to side 5 of the Σ child (child 4) belongs to child 7 (Table 2); so a move whose edge lies on the rim of child 1 keeps it at least two edges from both ends of its side.

Let C, C' and φ be as in the proof of Lemma 7.1. Both have the signature (by the proof of Proposition 7.4), and the junctions of C' are the images of those of C, since junctions are

vertices of children; so are the vertices V_j of child 1, and φ keeps the order of positions on its rim. For a port x of child 1 at an old vertex, the ribbon from $\varphi(x)$ reaches the rim at the image of the rim port reached from x, and $\varphi(x)$ lies in the same row of Table 6 as x. It remains to consider the new vertices v_1, v_2 when the edge $e = u_0u_1$ lies on the rim of child 1. Then v_1, v_2, u_0, u_1 lie strictly inside the same side of child 1, at least two edges from its ends, so in the same row of Table 6, and none is a crossover corner. If e lies on the rim of the composition (that is, on side 2 of child 1), v_1 and v_2 are hooks of C', as the table says. Otherwise, by the last part of the Zigzag Lemma (Lemma 5.1, with $n = 3$), the hooks of child 1 at v_1 and v_2 lie on the strands of C' through φ of the ports of child 1 at u_1 and u_0. Those ports lie in the same row, so these strands are arcs of C' with an end at the junction that the row names; an arc with an end at a hook is that hook, and both its ends are at the junction (junction q is a hook of C': C' has the signature, and junction q is at least one child side, so at least five edges, along the rim from junction 0, which is A_Γ, so it is neither A_Γ nor B_Γ). So the ribbons from the ports at v_1 and v_2 reach the rim there too. □

Lemma 8.8 (Where the junctions lie)**.** *Let $m \geq 1$. The junctions* 9, 12, 16, 17 *and* 21 *of N_{m+1} lie on its sides* 2, 3, 4, 4 *and* 5 *respectively, at least five edges from both ends of the side. Every rim vertex strictly inside side* 2 *of N_m lies on side* 4 *of N_{m+1}, at least five edges from both ends.*

Proof. By Table 3, side 2 of a Δ is made of the children's sides at positions 7 to 10, side 3 of those at 11 to 13, side 4 of those at 14 to 18, and side 5 of those at 19 to 21. Junction q lies between the sides at positions $q - 1$ and q, which lie in the same side of N_{m+1} for $q = 9, 12, 16, 17, 21$; and side 2 of N_m is the side at position 16. Every side of a supertile of level ≥ 1 has at least five edges (Lemma 6.5). □

Proof of Theorem 8.5. A rim vertex of N_m at least two edges from both ends of its side J is not a crossover corner, and by Lemma 8.7 its hook lies on the hook of N_{m+1} that Table 6 names for side J: for even m, junction 17, 17, 16, 12, 9 for $J = 0, 1, 3, 4, 5$; for odd m, junction 21, 17, 16, 16, 16; and for $J = 2$, the vertex is itself a hook of N_{m+1}, on its side 4. By Lemma 8.8, each of these hooks of N_{m+1} lies at least five edges inside a side of N_{m+1}, so the same rule applies to it one level up.

Let $b \geq 1$ be even, so that $\omega_b = +1$. By Lemma 8.7 every arc of N_b lies on H_{17}, H_{16}, H_{12} or H_9 of N_{b+1}, or on a hook of N_{b+1} strictly inside side 2 of N_b; by Lemma 8.8 these are hooks inside sides 4, 4, 3, 2 and 4 of N_{b+1}. Since $b + 1$ is odd, the hooks inside sides 3 and 4 lie on H_{16} of N_{b+2}, and those inside side 2 are hooks of N_{b+2} inside its side 4. All are inside side 4 of N_{b+2}, and since $b + 2$ is even they lie on H_{12} of N_{b+3}. So $\mathcal{P}_3(b)$ has a single part.

Let $b \geq 1$ be odd. Every arc of N_b lies on H_{21}, H_{17} or H_{16} of N_{b+1}, or on a hook of N_{b+1} strictly inside side 2 of N_b: inside sides 5, 4, 4, 4 of N_{b+1}. Since $b + 1$ is even, those inside side 4 lie on H_{12} and those inside side 5 on H_9 of N_{b+2}, inside its sides 3 and 2. Since $b + 2$ is odd, the first lie on H_{16} of N_{b+3}, and the second are hooks of N_{b+3} inside its side 4. All are inside side 4 of N_{b+3}, and since $b + 3$ is even they lie on H_{12} of N_{b+4}. So $\mathcal{P}_4(b)$ has a single part. Once $\mathcal{P}_k(b)$ has a single part, so does $\mathcal{P}_{k'}(b)$ for every $k' \geq k$.

The same steps give the last two rows of Table 5. For even b the arcs of N_b lie on two arcs of N_{b+2}: H_{16}, and the hook at junction 9 of N_{b+1}, which lies inside side 2 of N_{b+1} and so is not junction 16 of N_{b+2} (an end of that side). For odd b they lie on H_{12} and H_9 of N_{b+2}, and on H_{16} of N_{b+3} and the hook at junction 9 of N_{b+2}, which for the same reason is not junction 16 of N_{b+3}. Both arcs are reached in each case (for instance from vertices well inside sides 4 and 5 of N_b, or sides 3 and 0), and distinct hooks are distinct arcs. □

Remark 8.9 (Other checks of Theorem 8.5). Besides Computations 7.3 and 7.5, the proof of Theorem 8.5 uses only Computation 8.6. The following agree with it, and are not used. (1) Table 5 was computed on the actual blocks for $b \leq 6$, following every ribbon up to level 9, and for $b \leq 3$ by a second program in exact arithmetic up to level 7. (2) The junction map was checked directly on $C_\Delta(\ell_m, d_m, \pm 1)$ for $m \leq 4$, after each of the 190 single moves from (ℓ_1, d_1) on the Δ composition whose edge, if it lies on child 1, is at least two edges

from the ends of its side (with both signs), and on the actual blocks $N_1 \subset N_2$, $N_2 \subset N_3$, $N_3 \subset N_4$, built from the tiles in exact arithmetic. (3) A second route follows the arcs of N_b up the nest with *label states*: finite records of which hooks and crossover arcs of N_{b+k} carry which arcs of N_b (as sets of labels), advanced one level at a time on the skeleton with the reduced walk (Proposition 5.4) and Corollary 5.5, with every position written as an integer affine function of the eight letter lengths at level b. Run symbolically over 4 relative levels, once for each parity of b, its branch decisions (at each branch it takes the alternative that holds at $b = 2$, resp. $b = 1$) are 583 distinct linear conditions (*obligations*) on these lengths for even b and 564 for odd b; every one follows, for every $b \geq 1$, from Lemma 6.5 (every letter length at least 5, and $\delta_b - d_b \geq 4$, where $d_b = \zeta_b$ by Proposition 6.2 and Lemma 6.4) by an explicit certificate (one equality of each parity holds identically). A second implementation of this computation makes its comparisons differently (142 and 144 obligations, certified in the same way). This gives Theorem 8.5 and the last two rows of Table 5 again, for every $b \geq 1$.

8.3. **One strand.**

Proof of the Ariadne Theorem. By Lemma 3.4, every component of the strand graph $\mathcal{G}(\mathcal{N})$ is a cycle or a two-way infinite path, and there are no cycles, as shown after Proposition 8.4. It remains to show that $\mathcal{G}(\mathcal{N})$ is connected. Let p, q be pieces of tiles t, t' of $\mathcal{N}$. By Proposition 8.4 both tiles lie in some N_b with $b \geq 1$. Since N_b has no closed strand, p and q lie on arcs of N_b. By Theorem 8.5 these arcs lie on one arc of N_{b+4}, so p and q are on the same strand. The strand passes through every tile, since every tile has pieces. □

9. Scope and open questions

Other tilings. The Endless Strand Theorem holds for every homochiral tiling. The Ariadne Theorem is proved for the nest tiling, whose chain of supertiles is $\Delta \subset \Delta \subset \cdots$ with child number 1 at every step. Every tiling whose hierarchy covers the plane has an analogous chain $(K_m, c_m)_{m \geq 1}$: the level-m supertile X_m containing a fixed tile has kind K_m and is child c_m of X_{m+1}. The induction of Section 7 does not see the chain, but the merging of Section 8 does: its finite part depends on the sequence (K_m, c_m), and so may its outcome.

Remark 9.1 (Three strands: a computation, not a theorem). Let $K_m = \Theta$, $c_m = 2$ for odd m and $K_m = \Gamma$, $c_m = 7$ for even m (Table 1). A tiling with this chain is constructed as in Construction 8.1, and its hierarchy covers the plane (the argument of Lemma 8.3 and Proposition 8.4 applies, with Tables 2 and 3). Following the ribbons in exact arithmetic, the arcs of X_1 lie on exactly three arcs of X_m for $3 \leq m \leq 8$, and so do those of X_2 and X_3 for $5 \leq m \leq 8$; the label bookkeeping of Remark 8.9 along this chain agrees, and still gives three at $m = 20$. If this persists, the pieces of this tiling form exactly three strands, each two-way infinite by the Endless Strand Theorem. We have not proved this, and the paper does not use it. For the other alignment (Γ at odd levels, Θ at even ones) the arcs of X_1 lie on a single arc of X_4. Along 90,000 chains chosen at random step by step (30,000 for each $b \leq 3$: the kind of X_b, and then at each level the parent's kind and the child number, uniformly among those allowed by Table 1), the arcs of X_b came to lie on a single arc of X_m for some $m \leq 20$ in all but five. The label bookkeeping along a chain (`hierarchy/chains_label.js`) agrees with exact strand following on 60 such chains up to level 8. The programs are in `experiments/`, with logs.

Faults. Some homochiral tilings contain two or more infinite supertiles, by the usual compactness argument: a limit of copies of a tiling moved so that the origin is a vertex of a level-m supertile, for $m = 1, 2, \ldots$, has the origin on the boundary of a supertile of every level (its supertiles are the limits of those of the copies, by the uniqueness in Theorem 2.1(a)). This is not used in the proofs. The Endless Strand Theorem covers these tilings: no strand closes up, across a fault or anywhere else. The Ariadne Theorem says nothing about them.

Other drawings and designs. The theorems concern only the pairing of ports (Definition 3.1) and the straight-through rule. They therefore apply to any drawing that realises these, including drawings on the curved Spectre, whose tilings correspond to the homochiral tilings by Tile(1, 1) [18, §2]. The method uses the decoration only through the signatures of the tile and of the Mystic (Lemma 4.4), the position of the crossover, which fixes A (Lemma 6.4), the level-1 computations (Computations 7.3 and 7.5), Lemma 7.12 for the Endless Strand Theorem, and the junction map (Computation 8.6) for the Ariadne Theorem. For another decoration in which the tile has the signature at some edge, these computations can be run again, but some arguments need re-checking, not just re-running: Lemma 6.4, the choice of moves in Lemmas 7.1 and 7.2 (which uses that only the Σ child has its crossover corner inside a side), and Lemma 7.12 depend on where the crossover corners lie. The outcome may differ.

Open questions.

(1) Which homochiral tilings whose hierarchy covers the plane carry a single strand? Is the number of strands always finite, and which numbers occur?
(2) In a tiling with a fault, how many strands are there? The Endless Strand Theorem rules out closed strands there, but says nothing about their number.
(3) Which other decorations of Tile(1, 1) have no closed strand in any tiling, or a single strand in the nest tiling? The method applies to any decoration in which the tile has the signature at some edge (see above).

Appendix A. The computations

The first subsection lists the computations the proof uses; the others corroborate it. All programs and their logs are published with this paper at `https://cortado-j.github.io/SPECTRE_KNOT/`, folder `code/`. One command, `./run_all.sh`, re-runs everything the proof needs in about twelve minutes on a four-core machine, except the two comparisons with figures of [18], which need its PDF (not included) and run when the variable `SMKGS_PDF` names it; `./run_all.sh --all` adds the corroborating computations to level 9, and takes about an hour in all (with the memory of the JavaScript programs capped at 6.5 GB). The file `code/README.md` maps every statement of this paper to the script that checks it. (The folder `referee/` holds further checks, written by the AI system when reviewing its own work.) The Lean formalisation is published with the programs, in the folder `lean/`; its `build.sh` builds and checks it in about six minutes, and its `README.md` maps the statements of this paper to the formal ones. Its inputs are Tables 1 to 4 and the nine level-1 supertiles as glued patches of tiles, computed by `exact/exact.py`; its statement file (`lean/SpectreKnot/Statement.lean`), with the data files and the basic types, contains every definition that the formal theorems depend on. The formal proofs follow Sections 3 to 8, with these differences of detail: strands are modelled by two involutions on ports (Lemma 3.4); the formal proof of Lemma 7.1 shows that a move creates no closed strand through Proposition 5.4(b),(c), rather than directly; Proposition 5.4(c) is proved in the form used, that if every reduced walk reaches the rim there is no closed strand; in Computation 8.6 the kernel computes the second part, and the first is proved for every level from the signature of child 1; Lemma 8.8 is proved with two edges from the ends of a side, which is what its use needs; and Lemma 8.3 is used in form (b).

What the proof computes, and how. Besides the comparisons with figures of [18] (see *Checks of the composition pattern*), the proof uses two kinds of computation. The geometric computations on actual supertiles are the Mystic (Lemma 4.4) and the level-1 data (vertex positions, ℓ_1, d_1, the position of A, and the signature of the level-1 supertiles, Computation 7.5). They are done in exact arithmetic by `exact/exact.py`, in Python. It represents every vertex exactly as an element $a + b\rho + c\rho^2 + d\rho^3$ of $\mathbb{Z}[\rho]$, $\rho = e^{i\pi/6}$, with integers a, b, c, d, and every motion as $z \mapsto \rho^k z + t$ (or $\rho^k \bar{z} + t$ in reflected frames). Floating point is used only

to read off the sign of the area of a polygon whose area exceeds 8 in absolute value. The computations on compositions (Computations 7.3 and 8.6) use only Tables 1 to 3 and the integers ℓ_1, d_1. The program `hierarchy/compositions.py` builds each composition, follows the ribbon from every port, and checks the statement as worded. The Lean formalisation repeats both computations in its kernel, from the same tables, and re-checks Computation 7.5 on the level-1 supertiles exported by `exact.py` (in the table below, the Lean files named are in `lean/SpectreKnot/`). The tables as printed are compared with the pattern extracted in exact arithmetic by `second/check_exact_tables.py`, through a transcription that `second/check_tex.py` compares with the source of this paper. All the programs were written by the same AI system.

The JavaScript programs in `core/` and most of `hierarchy/` repeat the geometric computations independently and extend them: the patterns for $m \le 7$, the signature up to level 9, the merging for $b \le 6$ up to level 9. These extensions corroborate the proof but are not used in it. So do the skeleton walks and the symbolic run of the label states (Remarks 7.8 and 8.9), each done by two programs: the 18 skeleton walks by `hierarchy/induction.js` and `faults/faults.py`, the symbolic run by `hierarchy/certnest.js` and `second/nest2.py`, which takes the pattern from the tables of this paper and shares no code with the first. They work with integers only; the integers involved are below $2 \cdot 10^7$ in absolute value, far below 2^{53}, so the parts written in JavaScript are exact as well. The programs share no code with `exact.py`, although both implement the key-point construction of [18, App. A] and so the same table of placements. They work in double-precision floating point and identify points by rounding coordinates to 10^{-2} or 10^{-3}. We do not bound their rounding error. A separate check (`referee/overlap.js`), which builds the supertiles of levels 1 to 7 in the same way, finds distinct vertices at distance at least 0.89, and points that it identifies within 4×10^{-12} of each other; and up to level 9 every rim found has exactly E_m edges (Proposition 6.2) and every rim vertex has degree 2, which a wrongly identified point would break. All programs compose the ribbon pairings with Lemma 4.2 and nothing else. (In the programs the sign ω_m is called `beta`.)

Further checks of the signature. The signature was also computed directly from the tiles of built supertiles, in exact arithmetic at levels 1 to 4 and in floating point at levels 1 to 9, for all nine kinds; at levels 1, 2, 3 the rims have 44, 160 and 652 edges for Γ, and 46, 182 and 758 for the other kinds. The compositions $C_T(\ell_m, d_m, \pm 1)$ were built and followed port by port for $m \le 4$, and every single move from (ℓ_1, d_1) was checked to keep the signature; the moves chosen in the proof of Lemma 7.2 were followed back from levels 2 and 3 (`referee/chain_check.py`). For the skeleton route of Remark 7.8, the skeletons obtained from (ℓ_2, d_2), (ℓ_3, d_3) and $(\ell_1 + 2\ell_2, d_1 + 2d_2)$ are identical to those from (ℓ_1, d_1) with the same ω, and the 18 skeletons at $(\ell_1, d_1, \pm 1)$ coincide with those extracted from the built supertiles at every level from 2 to 9. As a check of Section 7.4, `faults/faults.py` computes, symbolically for every level $m \ge 1$ with certified obligations, every pair of supertiles glued along a side (264 ordered pairs) and every triple around a vertex (489 ordered stars) allowed by the matching rule. None of these unions contains a closed strand, except three Γ-supertiles around a point that is vertex 5 of each (m odd) or vertex 1 of each (m even). By the argument of Section 7.4, a closed strand there passes the corners A of at least two of the three (by the proof of the Endless Strand Theorem), each at distance β_m (m odd) or α_m (m even) from that point, so its crossing edges number at least $2\beta_m$, too many for Lemma 7.10 at level m, so it is excluded only at a higher level; by the Endless Strand Theorem this configuration does not occur in any homochiral tiling. Unions glued by letter but against the signs of the matching rule do contain closed strands. The patterns of the compositions $m \to m + 1$ for $m \le 3$ were extracted in exact arithmetic and agree with Tables 1 to 3; for $m = 4$ the kinds and gluing of the children agree with Tables 1 and 2.

Checks of the composition pattern. The comparison with [18, Fig. 4.1] in the proof of Proposition 2.4 works on a render of the figure. For each cluster the script finds the six dots, fits our six vertices to them by a proper similarity with residual below 1% of an edge (any other matching, or the vertices of any other kind, fits at least 40 times worse), finds our whole rim on the printed outline, every inner edge of our tiles on a printed line and every printed line inside the cluster on one of our edges, and our first Mystic tile on the lighter shaded tile, and compares the labels, read from the figure by hand, with Table 4.

For the check of Remark 2.5 against built supertiles, at levels $m \le 4$ we used the vertices observed as the points where three level-m supertiles meet inside larger supertiles. For $m = 5, \dots, 8$ we propagated them by the rule of Proposition 2.4(c), learnt at the observed level of the same parity; at levels 5 and 6, where vertices were also observed, the two agree (for Ψ, at the four of its six vertices that were observed). So for the compositions $m \to m+1$ with $m = 6, 7$ the rule (c) is an input of the extraction rather than something it tests. For the encoding of the panels in Remark 2.5, the kinds, positions and edge labels of the nine panels of [18, Fig. 5.1] were encoded by hand, and the marked vertices (with the centre of each panel's Γ hexagon) by a script from a render of the figure, which re-runs from the PDF. The script also confirms that the side-to-label dictionary (∗) is the same in all 71 placements of a child. Two deliberately corrupted encodings (two outer labels exchanged; one wrong label in [18, Fig. 4.2]) were rejected. We also checked that the tiles of every supertile of levels 1 to 7 (up to 2,147,679 tiles) have disjoint interiors and exactly fill the region bounded by its rim.

Further checks of the merging. The junction map (Table 6) was also checked on $C_\Delta(\ell_m, d_m, \pm 1)$ for $m \le 4$, after each of the 190 single moves of Remark 8.9(2) (`hierarchy/compositions.py`), and on the actual blocks $N_1 \subset N_2, N_2 \subset N_3, N_3 \subset N_4$ in exact arithmetic (`referee/junction_actual.py`), with a negative control.

In the label route of Remark 8.9(3), the symbolic run records 583 distinct obligations for even b and 564 for odd b; for each parity one is an equality, which holds identically, and 7 use $\delta_b - d_b \ge 4$. The second implementation, `second/nest2.py`, also works in the eight letter lengths at level b, with the same representatives $b = 2$ and $b = 1$ for the two parities, but makes its comparisons differently: it records 142 and 144 distinct obligations and certifies them in the same way. It reaches the same numbers of labelled arcs, checks at every step that the labels account for every arc of N_b exactly once, finds that the concrete runs for $1 \le b \le 40$ take the branches of the symbolic run of their parity, and rejects deliberately corrupted inputs and false obligations. The labels that the symbolic run places, and their grouping, agree at every relative level with a run rebuilt from the composition pattern, the parities and ω alone, and that run agrees with the same data read off the built blocks N_b, $b \le 6$, up to level 9. The merging for $b = 1$ was also found by following every ribbon of the built blocks in exact arithmetic, which gives 10, 2, 2, 1, and by the label computation with the actual positions (`hierarchy/nest_b1.js`). The other direct computations of the merging are described in the caption of Table 5.

Sizes. The script `hierarchy/sizes.py` computes these numbers from Table 1 and Proposition 6.2. They agree with the supertiles built by the programs (tile counts up to level 7, rim lengths up to level 9).

level m	1	2	3	4	5	6	7	8	9
tiles in a Δ	9	71	559	4,401	34,649	272,791	2,147,679	16,908,641	133,121,449
rim $E_m(\Delta)$	46	182	758	3,198	13,534	57,318	242,790	1,028,462	4,356,622
shortest side β_m	5	15	57	235	989	4,183	17,713	75,027	317,813

What was computed.

statement	computation	script
Definition 3.1 and Lemma 4.4	the tile's pairing; the Mystic, exact and in floating point	`core/tile.js`; `exact/exact.py`, `hierarchy/checks.js`
Proposition 2.4, Remark 2.5	vertices of levels 1–6 (points where three supertiles meet); the pattern of compositions $m \to m+1$, $m = 1, \ldots, 7$, all identical; exact for $m \le 3$	`hierarchy/vertices.js`, `combin.js`, `exact/exact.py`
Tables 1 to 4	the printed tables equal the patterns, side lengths and Σ offsets extracted in exact arithmetic (levels 1–4); children of the built supertiles of levels 2–5 glued as in Table 2	`second/check_exact_tables.py`, `second/check_tex.py`, `exact/exact.py`
Proposition 2.4 (base of $(*)$)	the level-1 supertiles as hat/turtle clusters, fitted to [18, Fig. 4.1] (dots, outline, inner tile edges, shading; labels read by hand)	`referee/fig41_compare.py`, `fig41_render.py`
Remark 2.5	agreement with an encoding of [18, Fig. 5.1], all nine kinds; its dots re-found from the PDF; two corrupted encodings rejected	`referee/fig51_compare.py`, `fig51_raster.py`
Remark 2.5	tiles of every supertile, levels 1–7, have disjoint interiors and fill the region bounded by its rim	`referee/overlap.js`
Propositions 6.1 and 6.2, Corollary 6.3, and Lemma 6.4	letters, recurrence, lengths at levels 1–8, closed forms	`hierarchy/recur.js`, `letters.py`
Computation 7.5	signature at level 1 (and directly at levels 2–4): exact, and in floating point	`exact/exact.py`, `hierarchy/checks.js`; Lean: `PatchSig.lean`
Computation 7.3	the 18 compositions $C_T(\ell_1, d_1, \pm 1)$ from the tables, every port followed; negative control	`hierarchy/compositions.py`; Lean: `CompFull.lean`
Computation 8.6	the junction map in $C_\Delta(\ell_1, d_1, \pm 1)$; negative control	`hierarchy/compositions.py`; Lean: `StepMap.lean`
Remark 7.8 (1), Remark 8.9 (2)	the compositions and the junction map at $(\ell_m, d_m, \pm 1)$, $m \le 4$; every single move from (ℓ_1, d_1)	`hierarchy/compositions.py`
Remark 7.8 (1)	the moves of Lemma 7.2 followed back from levels 2 and 3, checked step by step	`referee/chain_check.py`
Remark 8.9 (2)	Table 6 on the actual blocks $N_1, \ldots, N_4$, exact arithmetic; negative control	`referee/junction_actual.py`
Remark 7.8 (3)	the 18 skeleton walks, by two programs; skeletons depend only on parities; equal to the skeletons of the built supertiles at levels 2–9 (the computed skeletons record slightly more than Definition 5.3, e.g. rim-run parities)	`hierarchy/induction.js`, `faults/faults.py`
Proposition 5.4 (check)	reduced walk equals direct composition at all 1,424 parent ports, levels 1–8	`hierarchy/checks.js`
Remark 8.9 (3)	symbolic label computation in the letter lengths at level b, 583 + 564 distinct obligations, all certified by Lemma 6.5	`hierarchy/certnest.js`
Remark 8.9 (3)	second implementation: symbolic label computation, 142 + 144 distinct obligations, certified by Lemma 6.5; concrete runs for $1 \le b \le 40$ follow the same branches; Table 5 reproduced for $1 \le b \le 6$; corrupted inputs rejected	`second/nest2.py`
Theorem 8.5 ($b = 1$, corroboration)	the label computation for $N_1, \ldots, N_5$ with the actual positions: $10, 2, 2, 1$ parts	`hierarchy/nest_b1.js`
Theorem 8.5, Table 5	strand following in exact arithmetic for $b \le 3$ to level 7, in floating point for $b \le 6$ to level 9; label computation on inputs from the built blocks for $b \le 6$ to level 9, and the same inputs rebuilt from the pattern, the parities and ω (corroboration); row $b = 0$ on the drawing	`exact/exact.py`, `core/hier2.js`, `core/nest.js`, `hierarchy/synthnest.js`, `drawing/onetile.js`
corroboration	signature and skeleton periodicity at levels 1–9, all kinds; 0 closed strands at every level	`core/reduce.js`, `hierarchy/synth.js`

statement	computation	script
Section 7.4 (corroboration)	every pair of supertiles glued along a side and every triple around a vertex allowed by the matching rule, symbolically for all levels, certified: no closed strand except in three Γ-supertiles around their vertex 5 or 1, whose strand is too long for Lemma 7.10 at that level; configurations against the matching rule (closed strands found)	`faults/faults.py`
Section 3 (drawing)	the drawn ribbons realise the pairing, levels 1–3; shared edges glue e_i to e_j with $i + j$ odd, levels 1–4	`drawing/checkmodel.js`, `exact/exact.py`
Remark 3.6	the drawn pieces: containment, the single crossing, angles at the edges	`drawing/drawcheck.py`

References

1. Michael Baake and Uwe Grimm, *Aperiodic order. Volume 1: A mathematical invitation*, Encyclopedia of Mathematics and its Applications, vol. 149, Cambridge University Press, Cambridge, 2013.
2. George Bain, *Celtic art: The methods of construction*, William MacLellan, Glasgow, 1951.
3. bohemian-miser, *Spectre: playing with the spectre tiling*, `https://github.com/bohemian-miser/Spectre`, file `docs/FASS_THEOREM.md` (accessed 8 October 2026), 2026.
4. Joseph I. Cline, *Interlace patterns emerging in a Penrose-type Islamic design*, Proceedings of Bridges 2024: Mathematics, Art, Music, Architecture, Culture, 2024, pp. 163–170.
5. Peter R. Cromwell, *Celtic knotwork: mathematical art*, The Mathematical Intelligencer **15** (1993), no. 1, 36–47.
6. ———, *Knots and links*, Cambridge University Press, Cambridge, 2004.
7. Leonardo de Moura and Sebastian Ullrich, *The Lean 4 theorem prover and programming language*, Automated Deduction – CADE 28, Lecture Notes in Computer Science, vol. 12699, Springer, 2021, pp. 625–635.
8. Gwen Fisher and Blake Mellor, *On the topology of Celtic knot designs*, Bridges: Mathematical Connections in Art, Music, and Science (Winfield, Kansas), 2004, pp. 37–44.
9. Felix Flicker, Steven H. Simon, and S. A. Parameswaran, *Classical dimers on Penrose tilings*, Physical Review X **10** (2020), 011005.
10. Branko Grünbaum and G. C. Shephard, *Tilings and patterns*, second ed., Dover, 2016.
11. W. Patrick Hooper, *Truchet tilings and renormalization*, arXiv:1103.6046, 2011.
12. Slavik Jablan, Ljiljana Radović, Radmila Sazdanović, and Ana Zeković, *Knots in art*, Symmetry **4** (2012), no. 2, 302–328.
13. Craig S. Kaplan, *The spectre: a chiral aperiodic monotile*, `https://cs.uwaterloo.ca/~csk/spectre/`.
14. Stefan Pautze, *Space-filling, self-similar curves of regular pentagons, heptagons and other n-gons*, Proceedings of Bridges 2021: Mathematics, Art, Music, Architecture, Culture, 2021, pp. 157–164.
15. Gábor Pete, *Corner percolation on $\mathbb{Z}^2$ and the square root of 17*, Annals of Probability **36** (2008), no. 5, 1711–1747.
16. Shobhna Singh, Jerome Lloyd, and Felix Flicker, *Hamiltonian cycles on Ammann–Beenker tilings*, Physical Review X **14** (2024), 031005.
17. David Smith, Joseph Samuel Myers, Craig S. Kaplan, and Chaim Goodman-Strauss, *An aperiodic monotile*, Combinatorial Theory **4** (2024), no. 1, Paper No. 6, arXiv:2303.10798.
18. ———, *A chiral aperiodic monotile*, Combinatorial Theory **4** (2024), no. 2, Paper No. 13, arXiv:2305.17743.
19. Simon Tatham, *Finite-state transducers for substitution tilings*, arXiv:2512.16595, 2025.

Independent researcher
Email address: `justin.e.roughley@gmail.com`